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Introduction to
Data Science
(IT4142E)

Contents

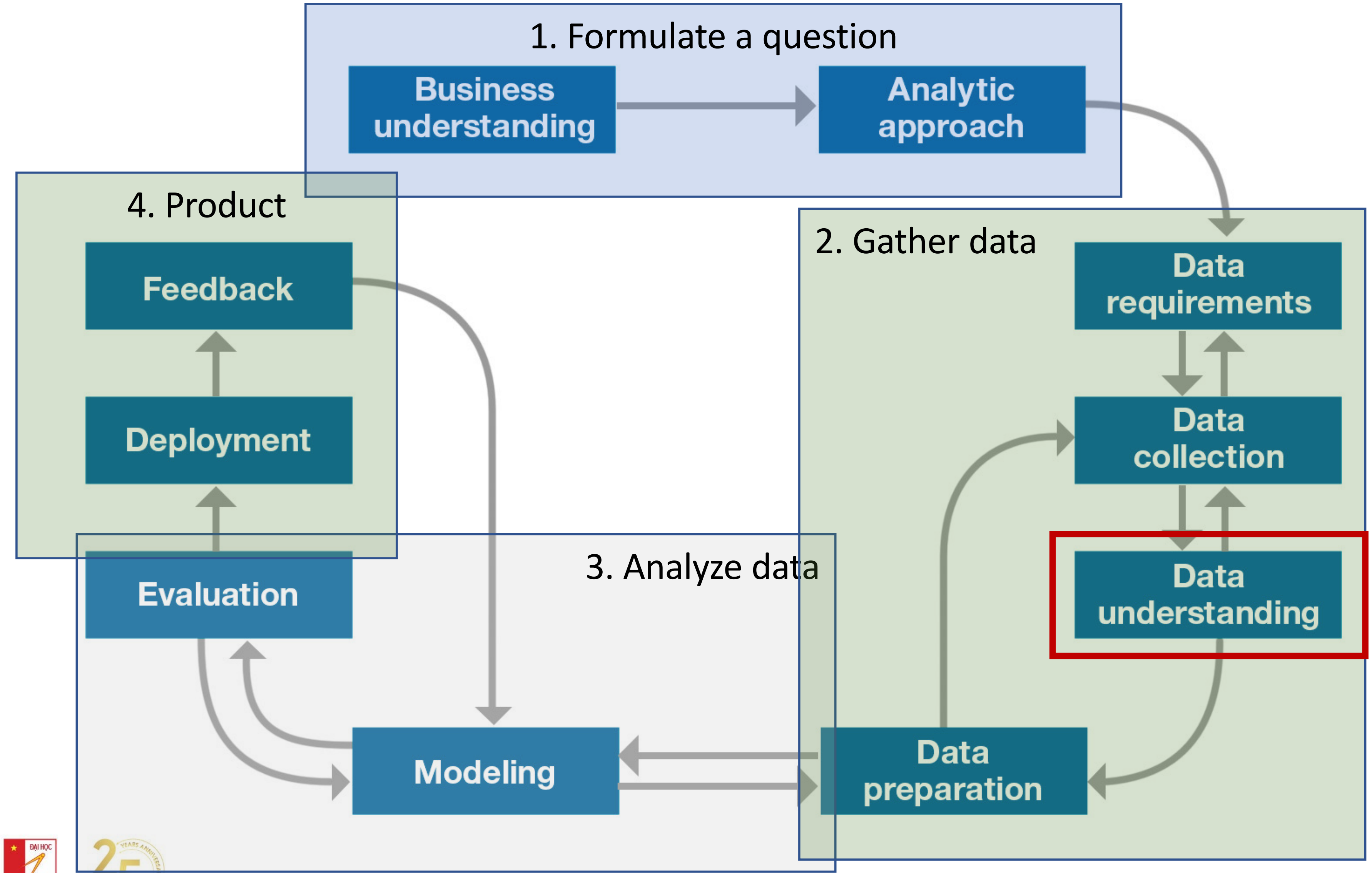
Learning outcomes

- Understand key elements in exploratory data analysis (EDA)
- Explain and use common summary statistics for EDA
- Plot and explain common graphs and charts for EDA

Motivation

- Before making inferences from data it is essential to examine all your variables.
  - To understand your data
- Why?
  - To listen to the data:
  - to catch mistakes
  - to see patterns in the data
  - to find violations of statistical assumptions
  - to generate hypotheses
  - …and because if you don’t, you will have trouble later

Data science process



Source: *Foundational Methodology for Data Science*, IBM, 2015

Exploratory data analysis (EDA) focus

- The focus is on the data—its structure, outliers, and models suggested by the data.
- EDA approach makes use of (and shows) all of the available data. In this sense there is no corresponding loss of information.
  - Summary statistics
  - Visualization
  - Clustering and anomaly detection
  - Dimensionality reduction

EDA definition

- The EDA is precisely not a set of techniques, but an attitude/philosophy about how a data analysis should be carried out.
  - Helps to select the right tool for preprocessing or analysis
  - Makes use of humans' abilities to recognize patterns in data

EDA common questions

- What is a typical value?
- What is the uncertainty for a typical value?
- What is a good distributional fit for a set of numbers?
- Does an engineering modification have an effect?
- Does a factor have an effect?
- What are the most important factors?
- Are measurements coming from different laboratories equivalent?
- What is the best function for relating a response variable to a set of factor variables?
- What are the best settings for factors?
- Can we separate signal from noise in time dependent data?
- Can we extract any structure from multivariate data?
- Does the data have outliers?

EDA is an iterative process

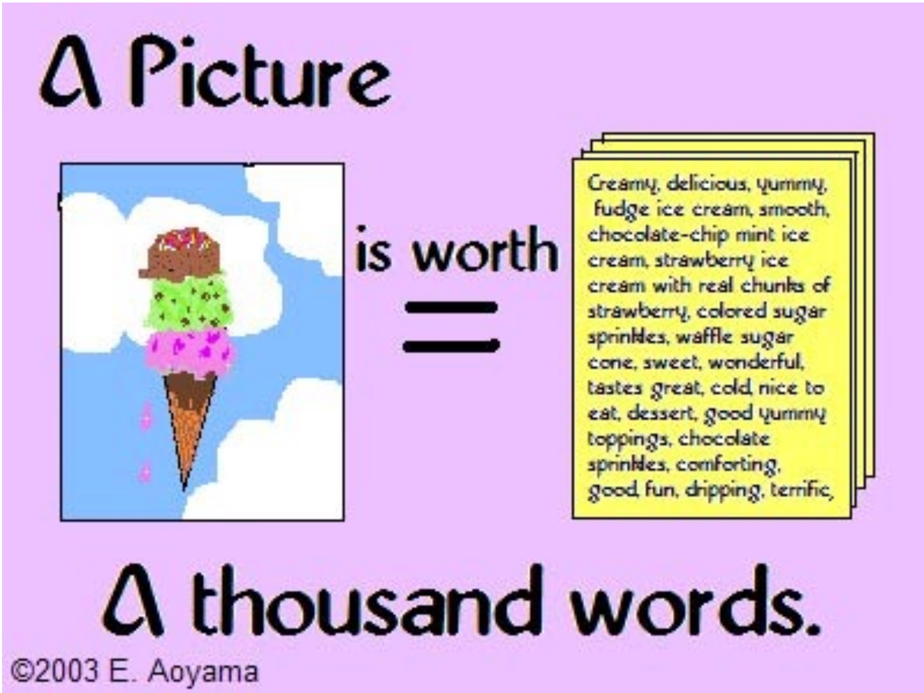
- **Repeat...**
  - Identify and prioritize relevant questions in decreasing order of importance
  - Ask questions
  - Construct graphics to address questions
  - Inspect “answer” and derive new questions

EDA strategy

- Examine variables one by one, then look at the relationships among the different variables
- Start with graphs, then add numerical summaries of specific aspects of the data
- Be aware of attribute types
  - Categorical vs. Numeric

EDA techniques

- Graphical techniques
  - scatter plots, character plots, box plots, histograms, probability plots, residual plots, and mean plots.
- Quantitative techniques



Describing univariate data

Observations and variables

feature

observations

- Data is an collection of observations
- an attribute is thought of as a set of values describing some aspect across all observations, it is called a variable

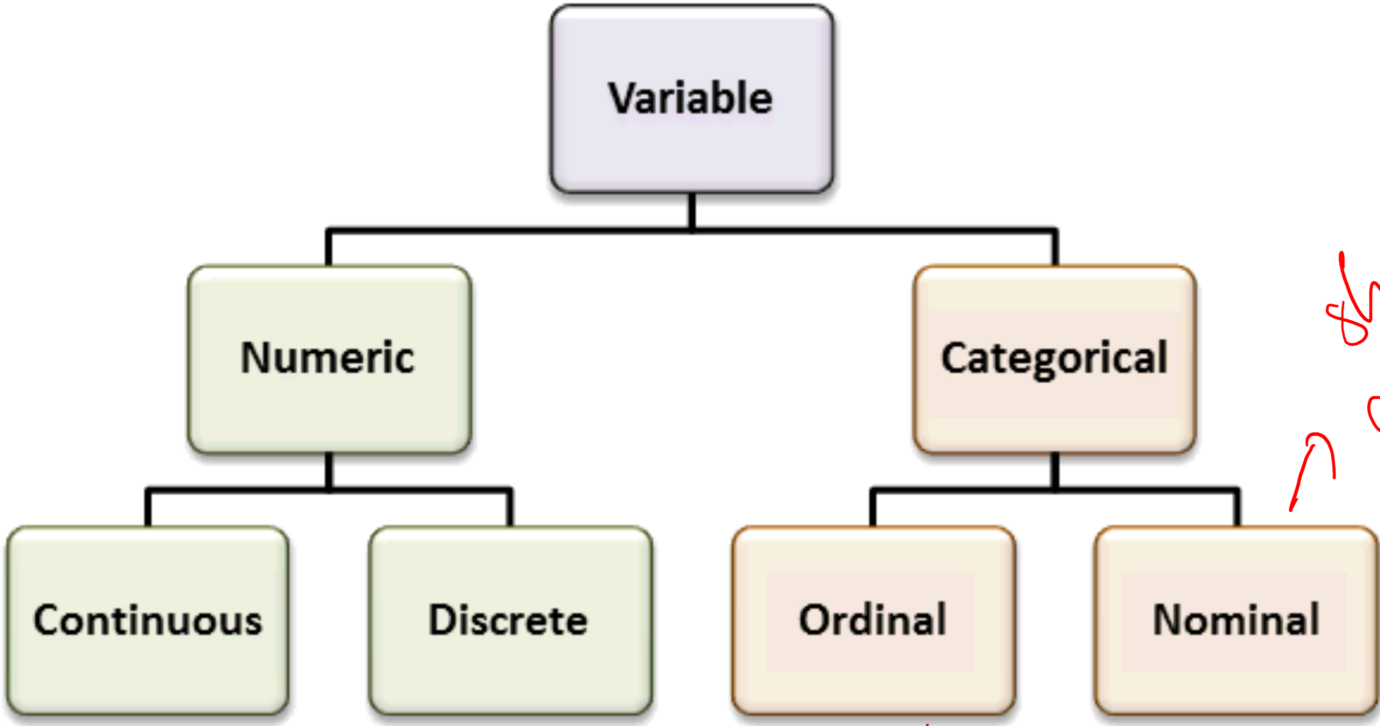
Variables / Features / Characteristics

HR Information		Contact	
Position	Salary	Office	Extn.
Accountant	$162,700	Tokyo	5407
Chief Executive Officer (CEO)	$1,200,000	London	5797
Junior Technical Author	$86,000	San Francisco	1562
Software Engineer	$132,000	London	2558

sample / observation

record / example / data point

Types of variables



shape: → circle
rectangle
square

size of clothes
s < m < l < ~~x~~l < xxl

Dimensionality of data sets

- Univariate: Measurement made on one variable per subject
- Bivariate: Measurement made on two variables per subject
- Multivariate: Measurement made on many variables per subject

Explore statistic information of data

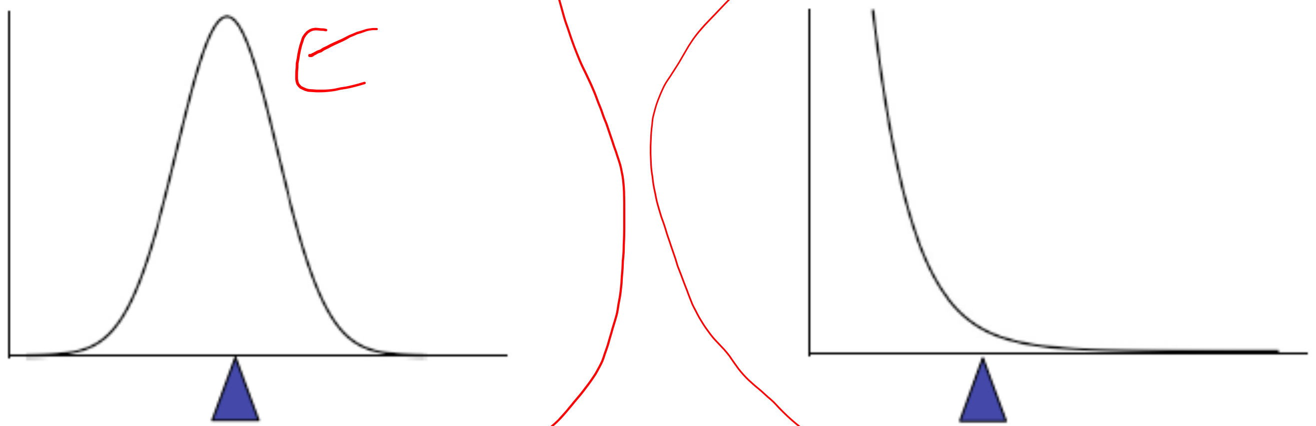
Measures of central tendency

- Measures of Location: estimate a location parameter for the distribution; i.e., to find a typical or central value that best describes the data.
- Measures of Scale: characterize the spread, or variability, of a data set. Measures of scale are simply attempts to estimate this variability.
- Skewness and Kurtosis

Mean

- To calculate the average value of a set of observations, sum of their values divided by the number of observations:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$



Median

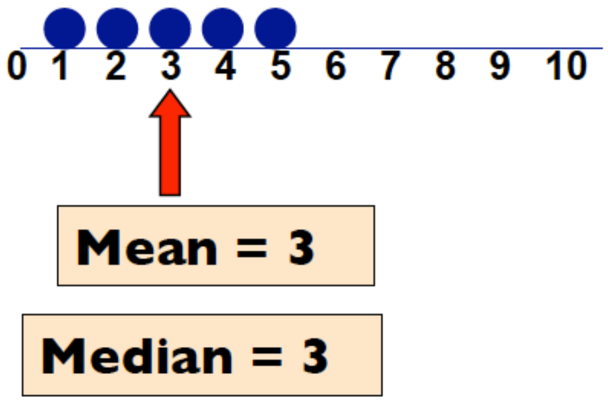
- The median is the value of the point which has half the data smaller than that point and half the data larger than that point.
- Calculation
  - If there are an odd number of observations, find the middle value
  - If there are an even number of observations, find the middle two values and average them
- Example
  - Age of participants: 17 19 21 22 23 23 23 38
  - **Median = (22+23)/2 = 22.5**

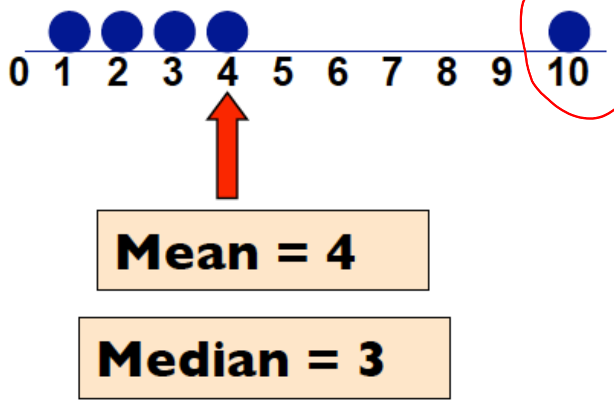
Mode

- mode is the most commonly reported value for a particular variable
  - Eg. 3, 4, 5, 6, 7, 7, 7, 8, 8, 9. Mode = 7
  - Eg. 3, 4, 5, 6, 7, 7, 7, 8, 8, 8, 9. Mode = {7, 8} = 7.5

Which location measure is best?

- Mean is best for symmetric distributions without outliers
- Median is useful for skewed distributions or data with outliers





Measure of scale : Variance and standard deviation

- Variance: average of squared deviations of values from the mean

f1   f2

[0, ... 10]   [1000, ... , 10,000]

$$\hat{\sigma}^2 = \frac{\sum_{i}^{n}(x_i - \bar{x})^2}{n-1}$$

- Standard Deviation: simply the square root of the variance

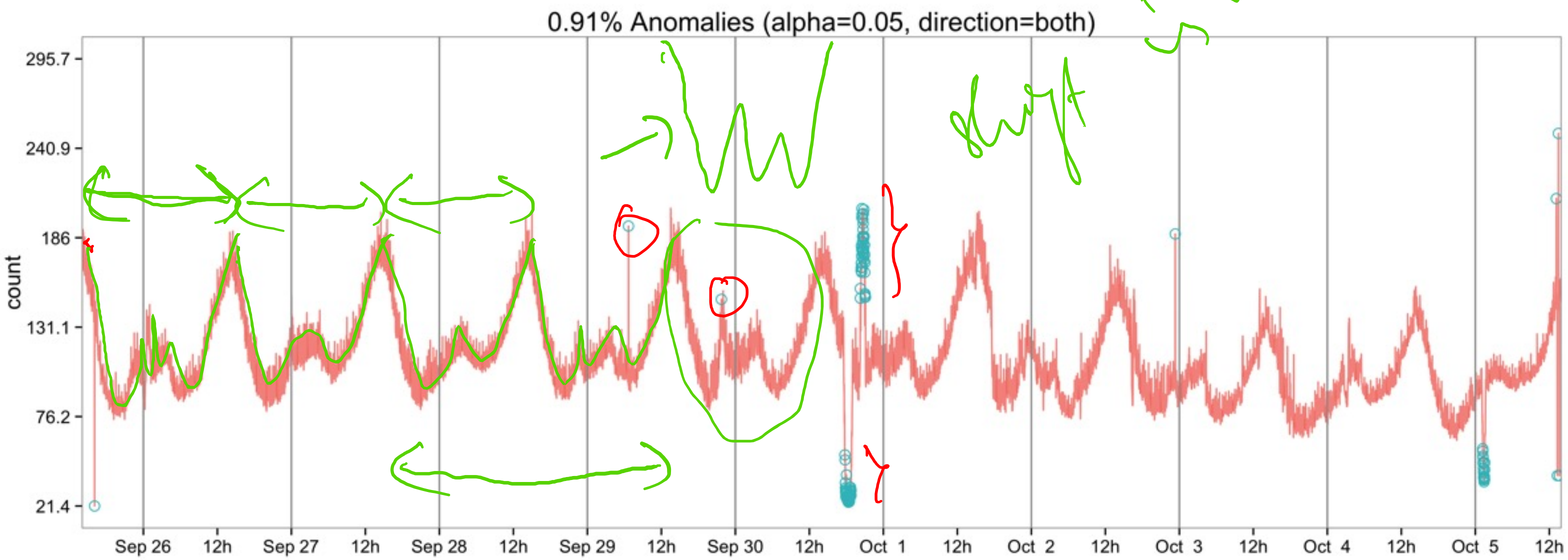
$$\hat{\sigma} = \sqrt{\frac{\sum_{i}^{n}(x_i - \bar{x})^2}{n-1}}$$

Explore data trends with charts

Run sequence plot

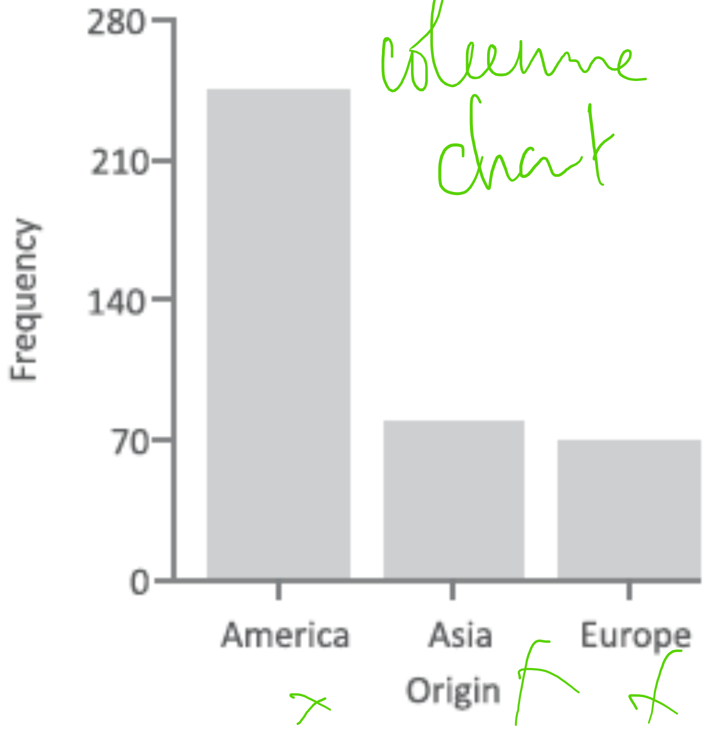
Time _ series

- displays observed data in a time sequence.
- The run sequence plot can be used to answer the following questions
  - Are there any shifts in location?
  - Are there any shifts in variation?
  - Are there any outliers?



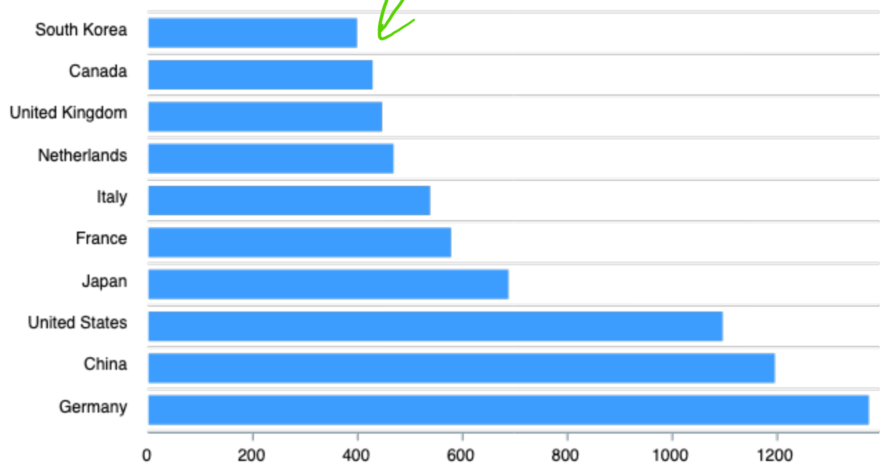
Bar charts

- a bar chart displays the relative frequencies for the different values.
- or a chart presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent



column chart

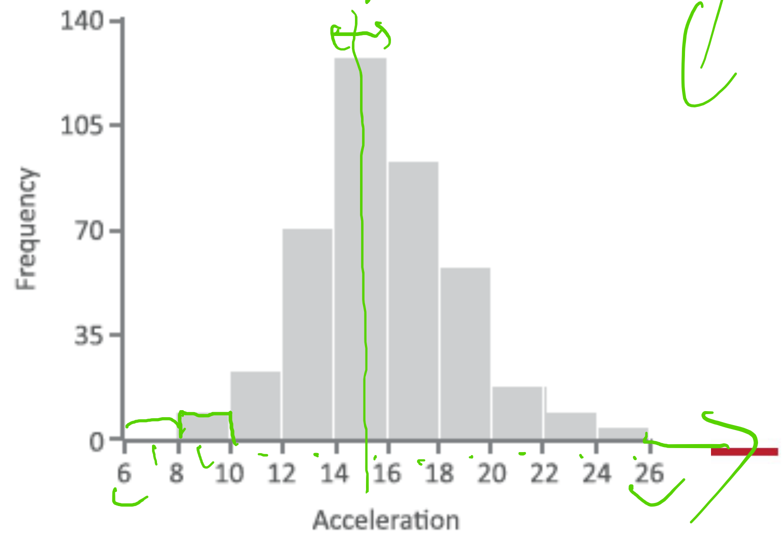
bar chart



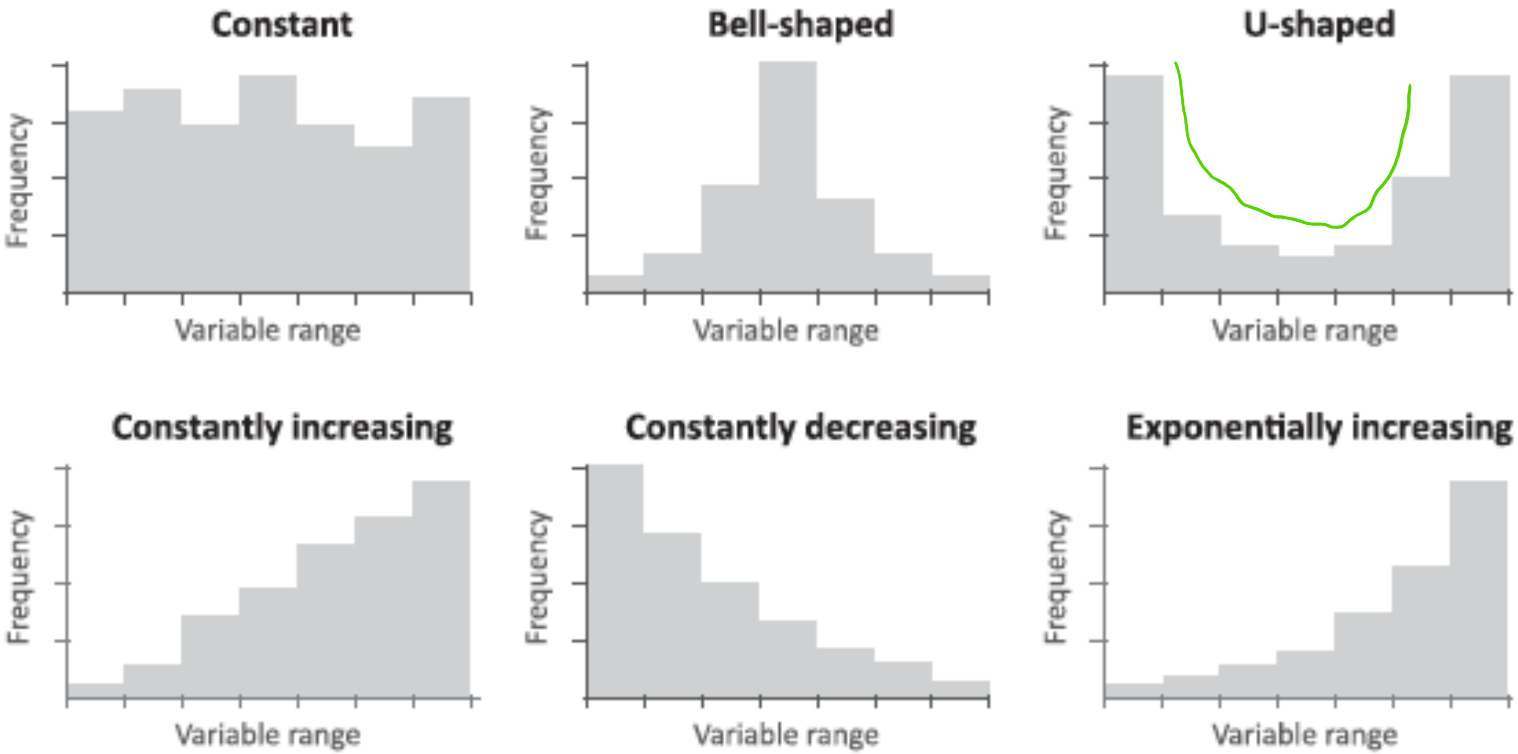
Histogram plot

- A histogram is to graphically summarize the distribution of a univariate data set.
- The histogram can be used to answer the following questions:
  - What kind of population distribution do the data come from?
  - Where are the data located?
  - How spread out are the data?
  - Are the data symmetric or skewed?
  - Are there outliers in the data?

tes bin

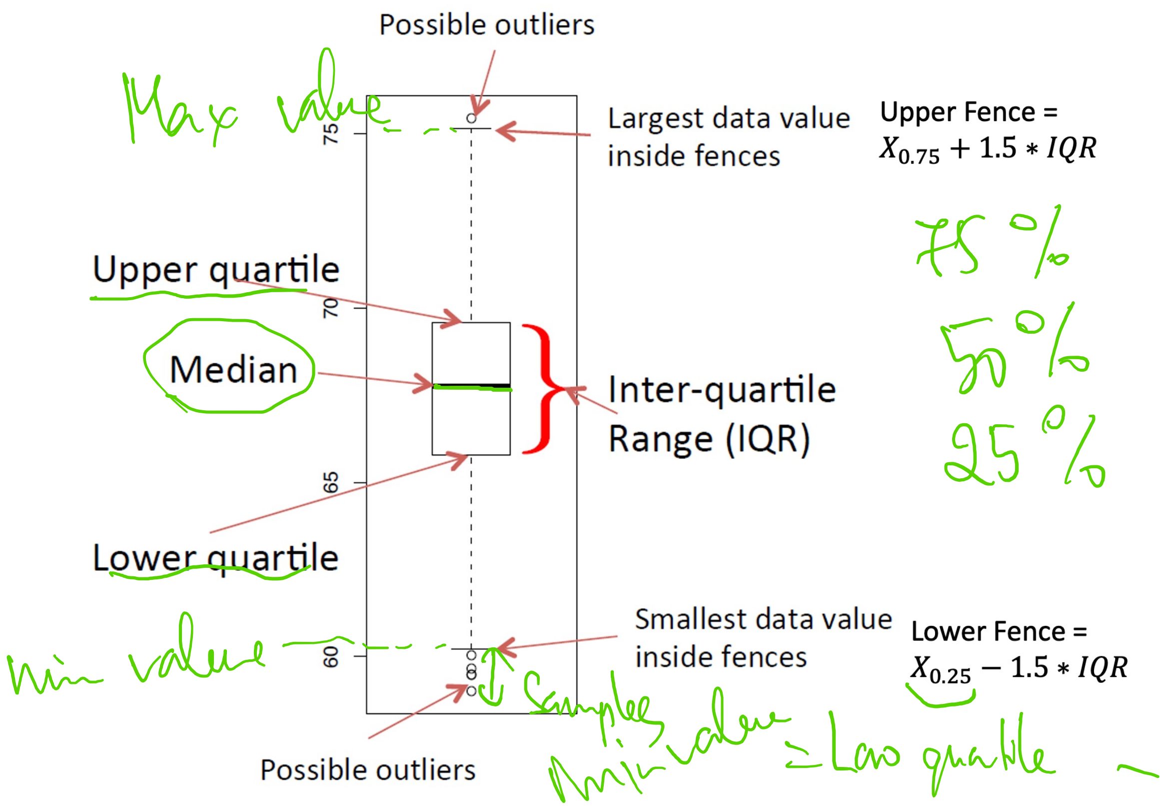


Example of frequency distributions



Box plot / violin plot (variant of box plot)

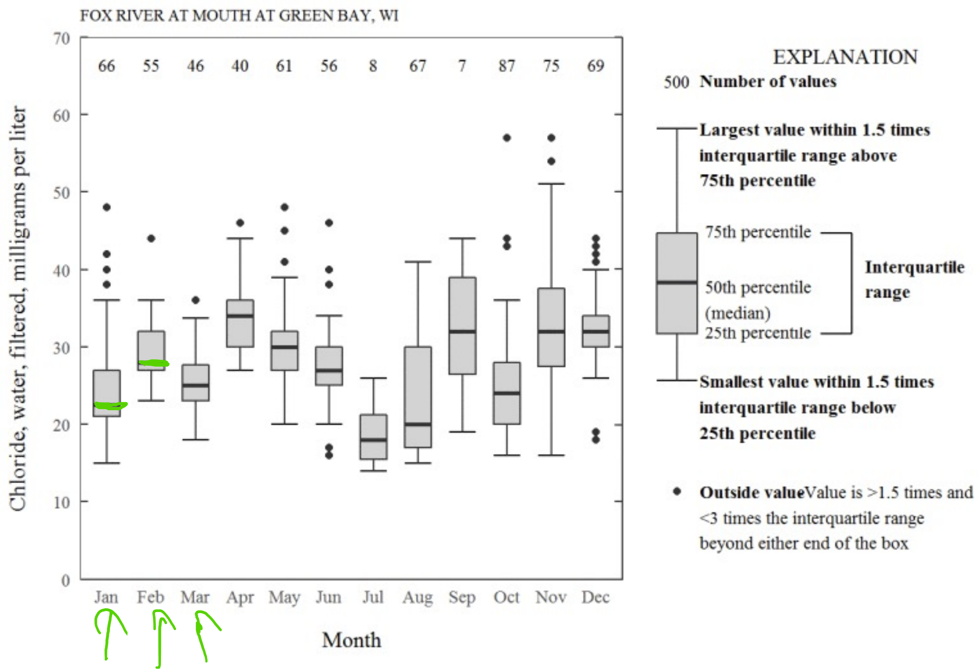
- Box plot displayed: the lowest value, the lower quartile (Q1), the median (Q2), the upper quartile (Q3), the highest value, and the mean.



Min value = Low quartile − 1,5 * IQR

Box plot (2)

- The box plot can provide answers to the following questions:
  - Is a factor significant?
  - Does the location differ between subgroups?
  - Does the variation differ between subgroups?
  - Are there any outliers?



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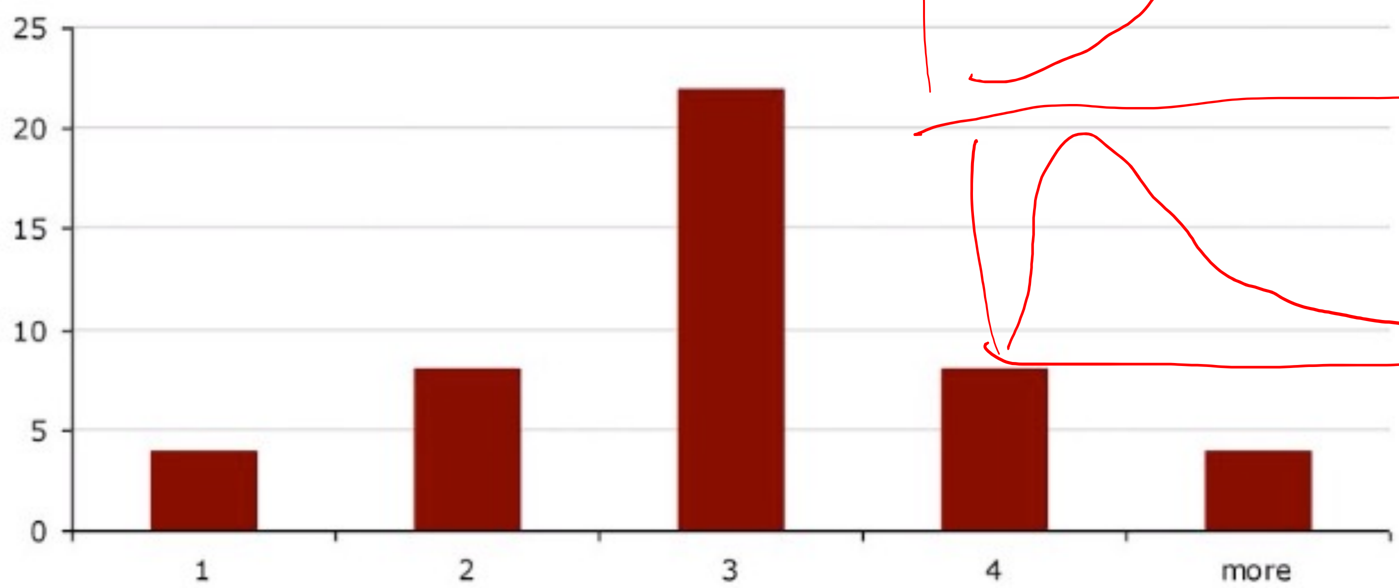
Skewness

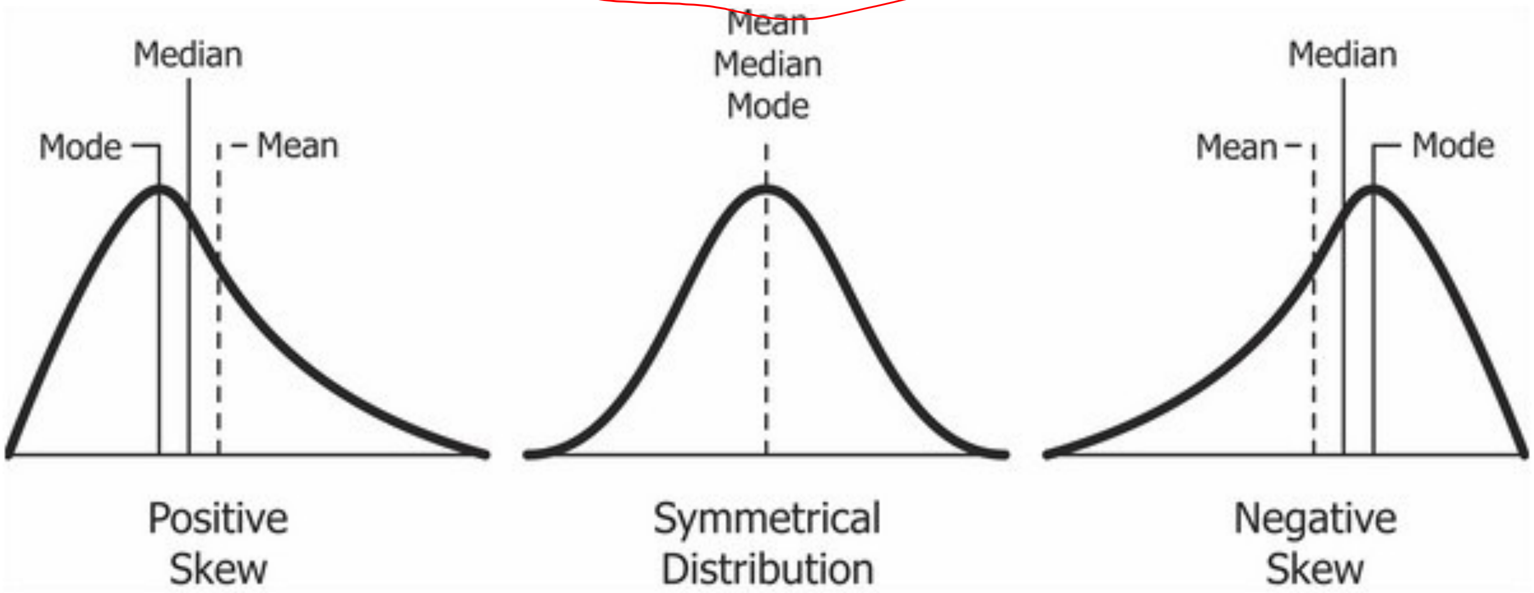
- Skewness is a measure of asymmetry. A distribution, or data set, is symmetric if it looks the same to the left and right of the center point
- Symetrical distribution



Mean = median = mode = 3

Negative, positive skewness

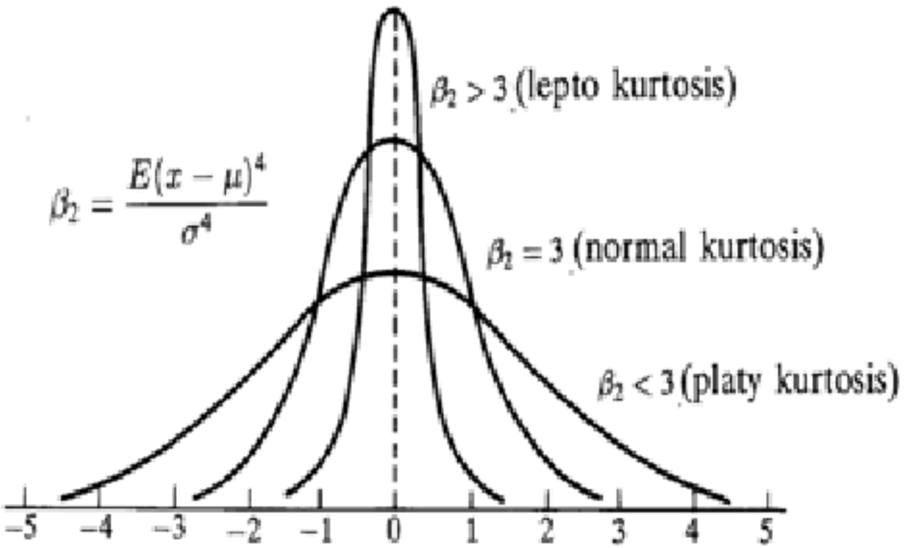
$$s_k = \frac{\sum_{i=1}^{T}(x_i - \bar{x})^3}{\sigma^3}$$



Kurtosis

- Kurtosis is a measure of whether the data are peaked or flat relative to a normal distribution. data sets with high kurtosis tend to have a distinct peak near the mean, decline rather rapidly, and have heavy tails. Data sets with low kurtosis tend to have a flat top near the mean rather than a sharp peak.

$$k = \frac{\sum_{i=1}^{T} (x_i - \bar{x})^4}{\sigma^4}$$



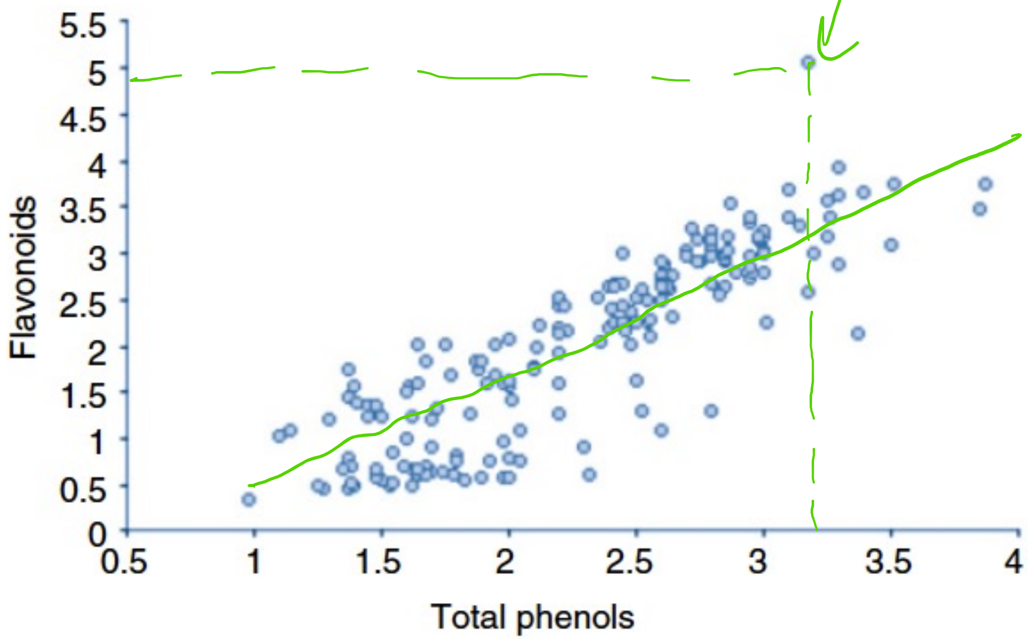
Understanding relationships

Scatter plot

- identify whether a relationship exists between two continuous variables measured on the ratio or interval scales
  - two variables are plotted on the x-and y-axis
  - each point is a single observation.

a sample

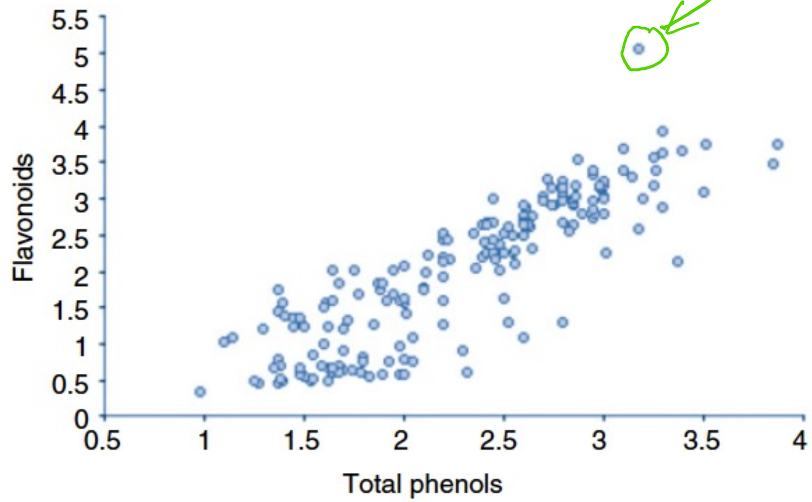
projection $\mathbb{R}^n \to \mathbb{R}^2$



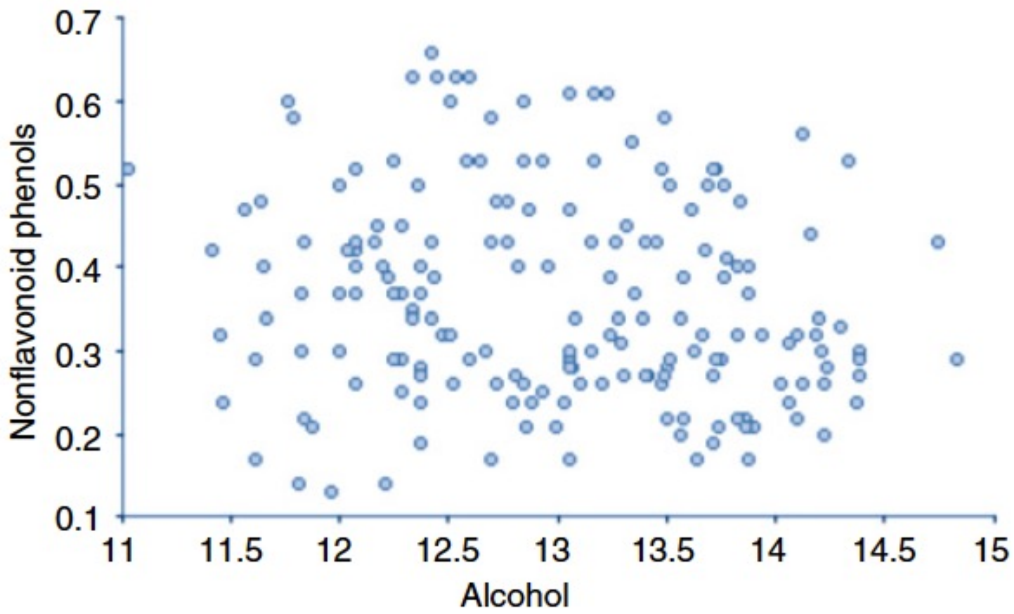
Scatter plot

- Scatter plots can provide answers to the following questions:
  - Are variables X and Y related?
  - Are variables X and Y linearly related?
  - Are variables X and Y non-linearly related?
  - Does the variation in Y change depending on X?
  - Are there outliers?

outlier

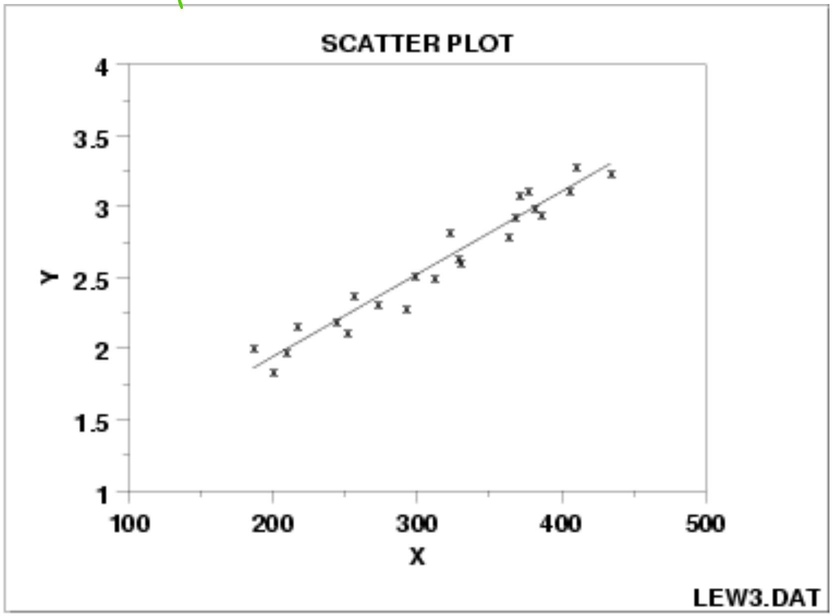


Scatter plot: No relationship

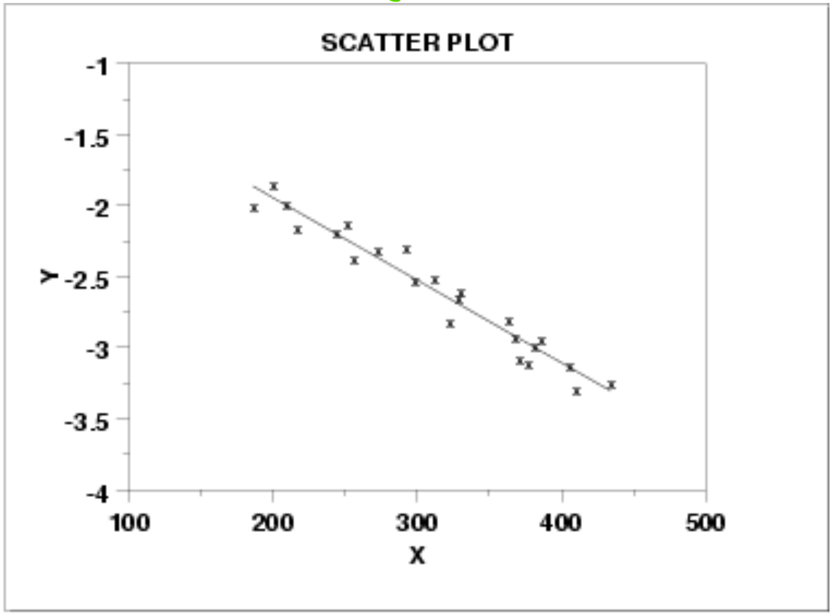


Scatter plot: Strong linear (positive - negative correlation)

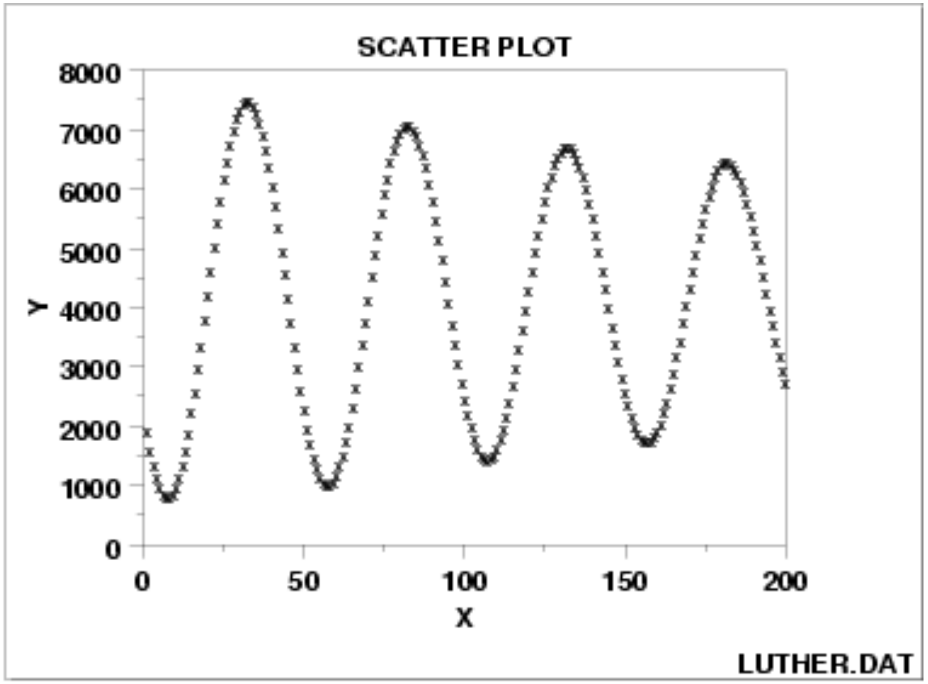
positive correlation



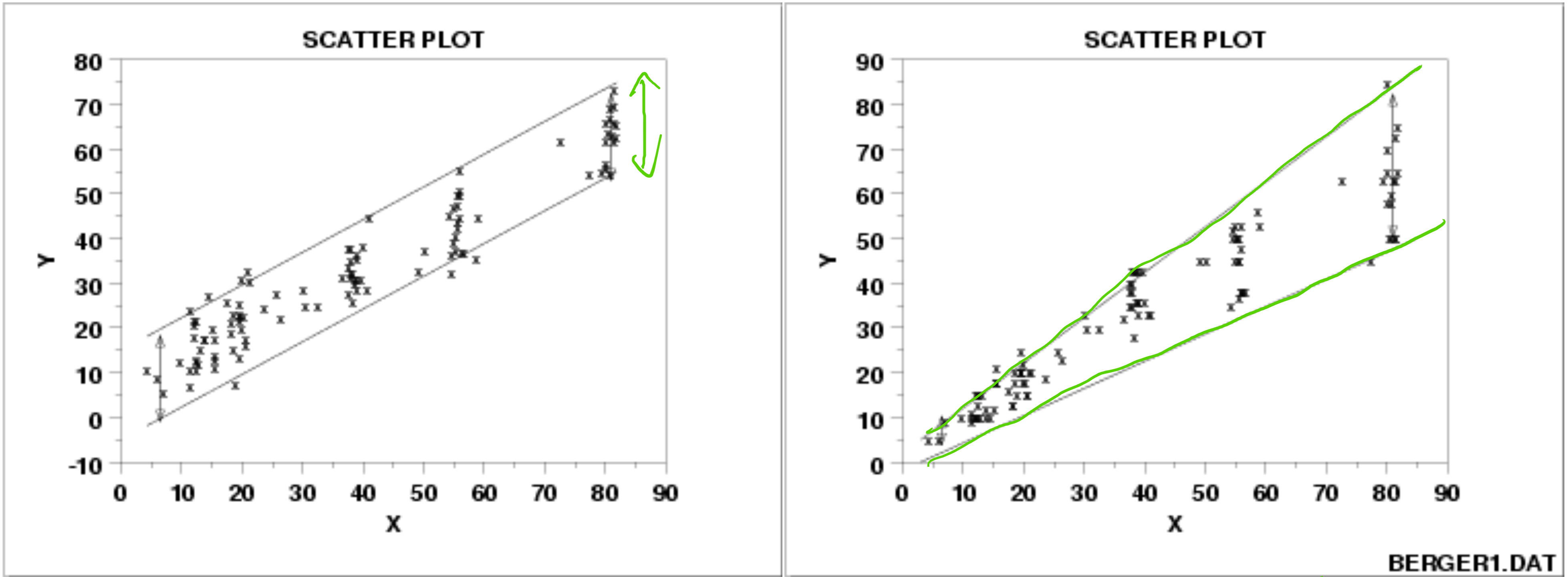
negative correlation



Scatter plot: Sinusoidal relationship (damped)

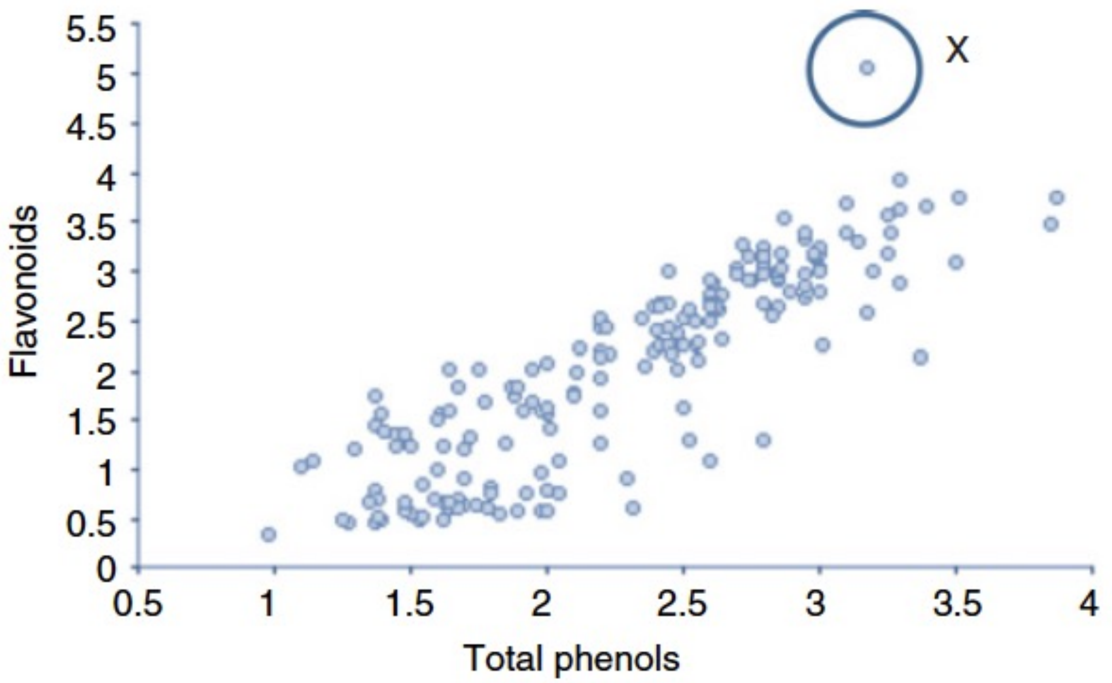


Scatter plot: variation of Y does not depend on X (homoscedastic)



x increase
variation of y increase too

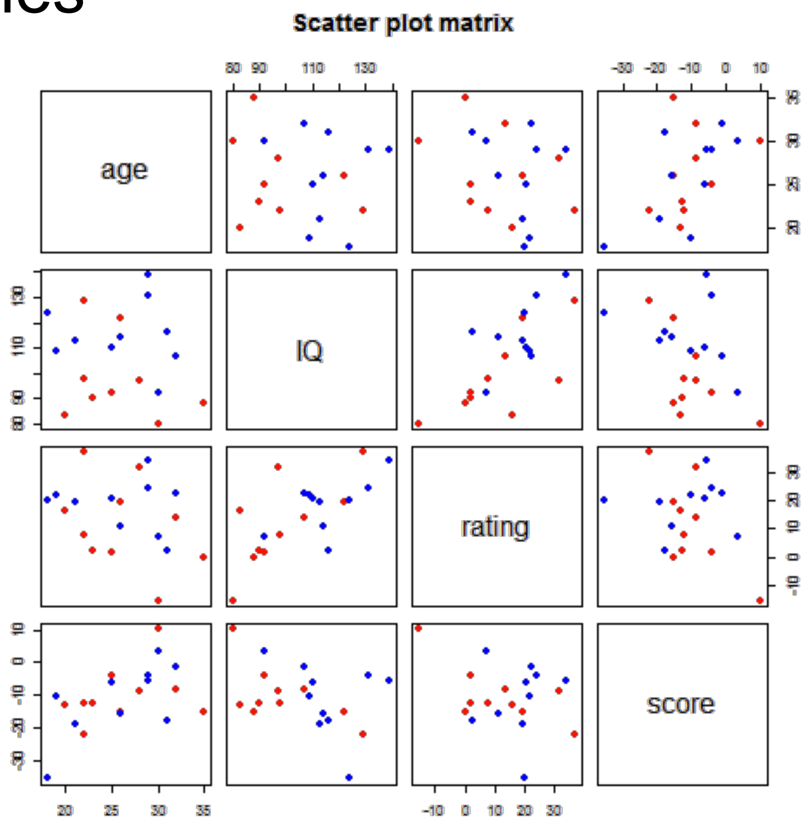
Scatter plot: Outlier



Scatterplot matrix

n features

- a collection of **scatterplots** organized into a grid (or **matrix**).
- Each **scatterplot** shows the relationship between a pair of variables

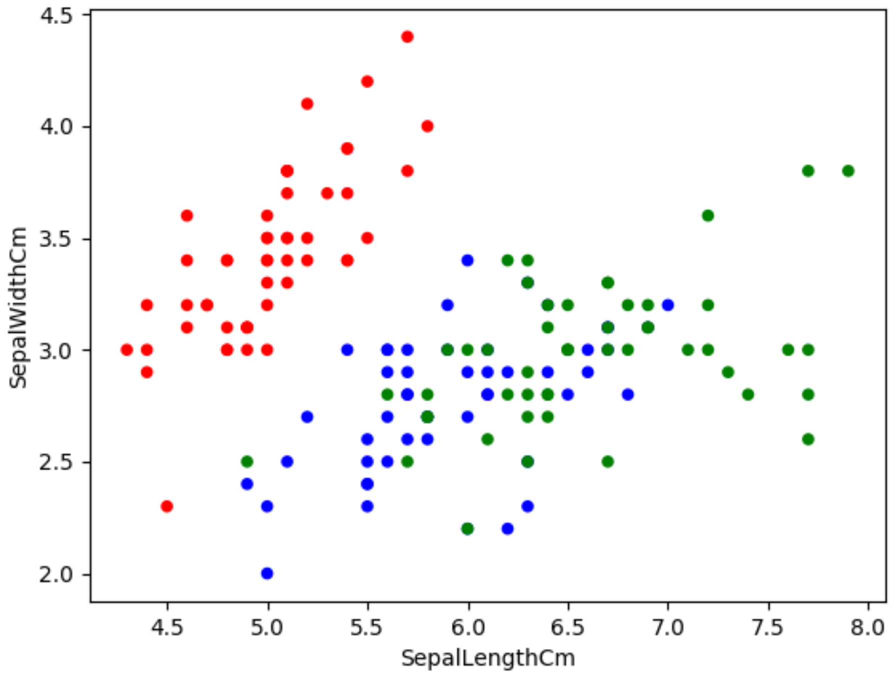


4 features

C^2_4 scatter plots

Use scatter plot for Iris data

- Display color for each kind of Iris




```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

iris =
pd.read_csv("../input/Iris.csv")
iris.head()

iris["Species"].value_counts()
col = iris['Species'].map({"Iris-
setosa":'r',     "Iris-
virginica":'g', "Iris-
versicolor":'b'})
iris.plot(kind="scatter",
x="SepalLengthCm",
y="SepalWidthCm", c=col)

plt.show()
```

Identifing and understanding groups

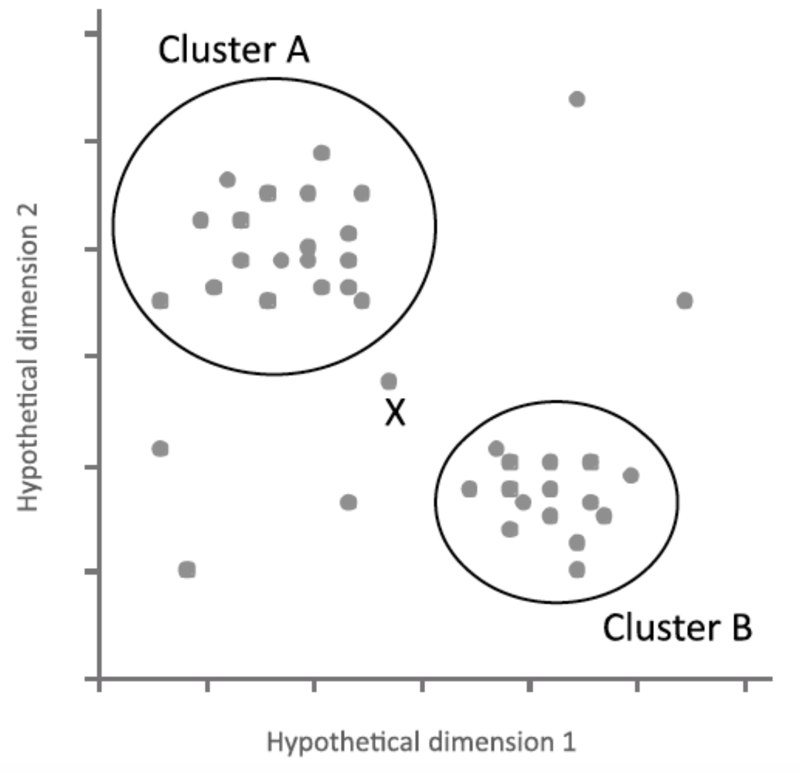
Clustering Methods in Exploratory Analysis

Motivation

- Decomposing a data set into simpler subsets helps make sense of the entire collection of observations
  - uncover relationships in the data such as groups of consumers who buy certain combinations of products
  - identify rules from the data
  - discover observations dissimilar from those in the major identified groups (possible errors or anomalies)

Clustering

- A way of grouping together data samples that are ***similar*** in some way - according to some criteria
- A form of ***unsupervised learning*** – you generally don't have examples demonstrating how the data *should* be grouped together



Can we find things that are close together?

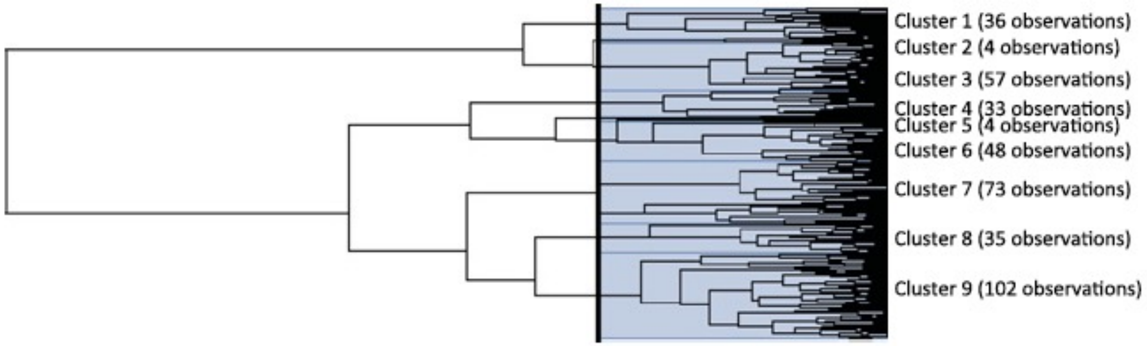
- Clustering organizes things that are close into groups
  - How do we define close?
  - How do we group things?
  - How do we visualize the grouping?
  - How do we interpret the grouping?

Types of clustering

- Hierarchical clustering
- Flat clustering

Hierarchical clustering

- An agglomerative approach
  - Find closest two things
  - Put them together
  - Find next closest
- Requires
  - A defined distance
  - A merging approach
- Produces
  - A tree showing how close things are to each other (dendrogram)



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Distances

- A method of clustering needs a way to measure how similar observations are to each other.
- Continuous - Euclidean distance
- Continuous - correlation similarity
- Binary - Manhattan distance
- Pick a distance/similarity that makes sense for the problem

Euclidean distance

$$d = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$$

ID	Variable 1	Variable 2	Variable 3	Variable 4	Variable 5
A	0.7	0.8	0.4	0.5	0.2
B	0.6	0.8	0.5	0.4	0.2
C	0.8	0.9	0.7	0.8	0.9

$$d_{\mathrm{A-B}} = \sqrt{(0.7-0.6)^2 + (0.8-0.8)^2 + (0.4-0.5)^2 + (0.5-0.4)^2 + (0.2-0.2)^2}$$
$$d_{\mathrm{A-B}} = 0.17$$

$$d_{\mathrm{A-C}} = \sqrt{(0.7-0.8)^2 + (0.8-0.9)^2 + (0.4-0.7)^2 + (0.5-0.8)^2 + (0.2-0.9)^2}$$
$$d_{\mathrm{A-C}} = 0.83$$

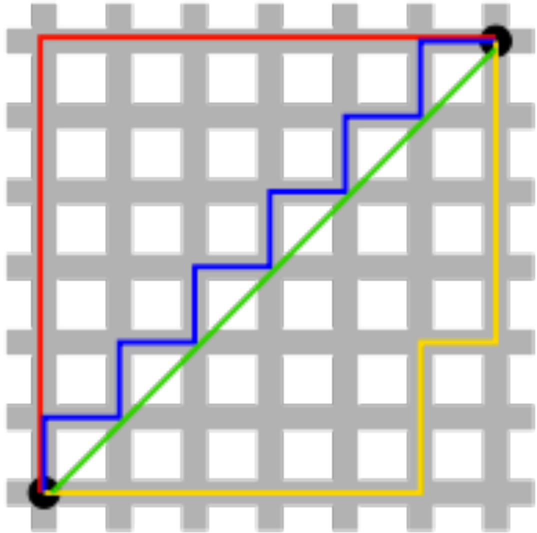
Manhattan distance

- is the sum of the lengths of the projections of the line segment between the points onto tl

$$d_1(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_1 = \sum_{i=1}^{n} |p_i - q_i|,$$

where $(\mathbf{p}, \mathbf{q})$ are vectors

$$\mathbf{p} = (p_1, p_2, \dots, p_n) \text{ and } \mathbf{q} = (q_1, q_2, \dots, q_n)$$

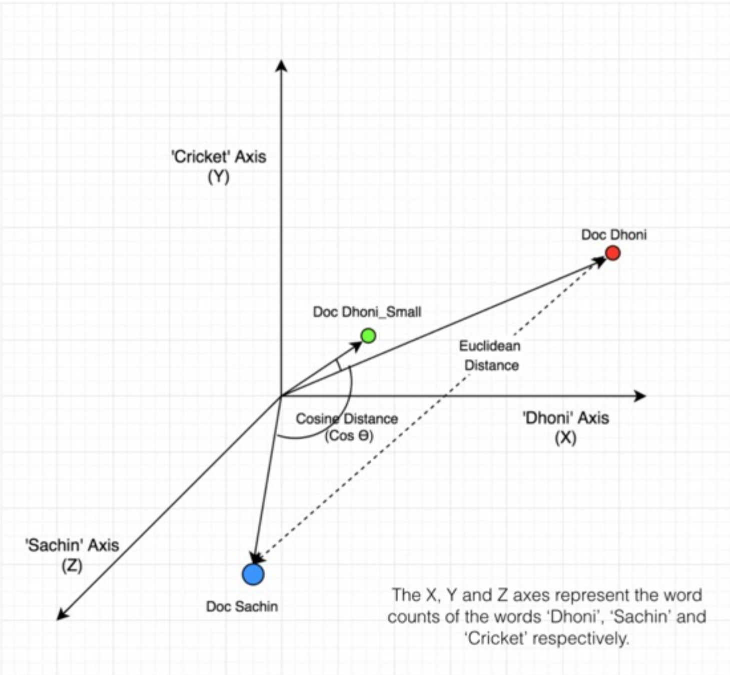


Cosine distance

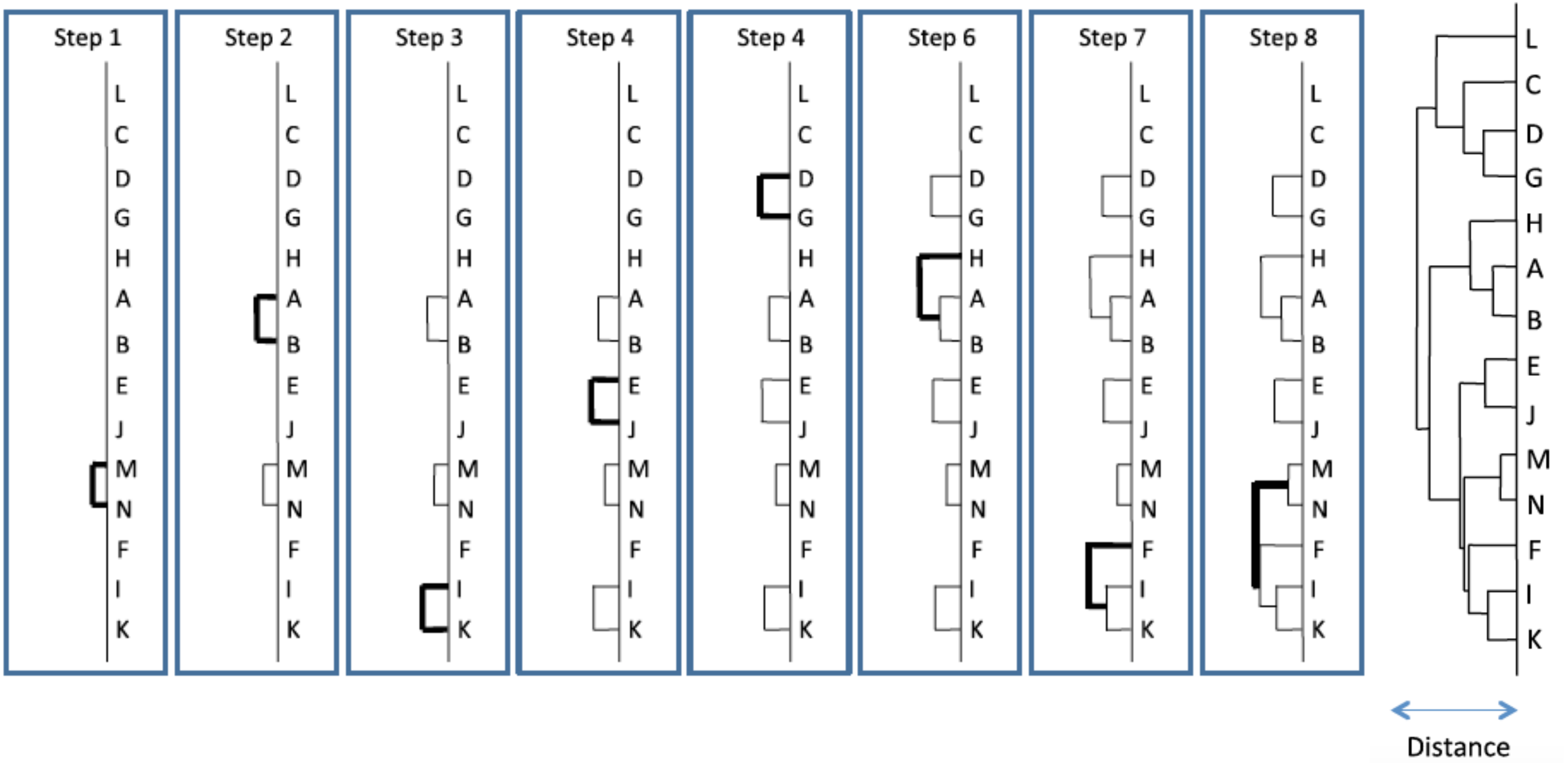
$$Cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \, \|\vec{b}\|} = \frac{\sum_1^n a_i b_i}{\sqrt{\sum_1^n a_i^2} \sqrt{\sum_1^n b_i^2}}$$

where, $\vec{a} \cdot \vec{b} = \sum_1^n a_i b_i = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$ is the dot product of the two vectors.

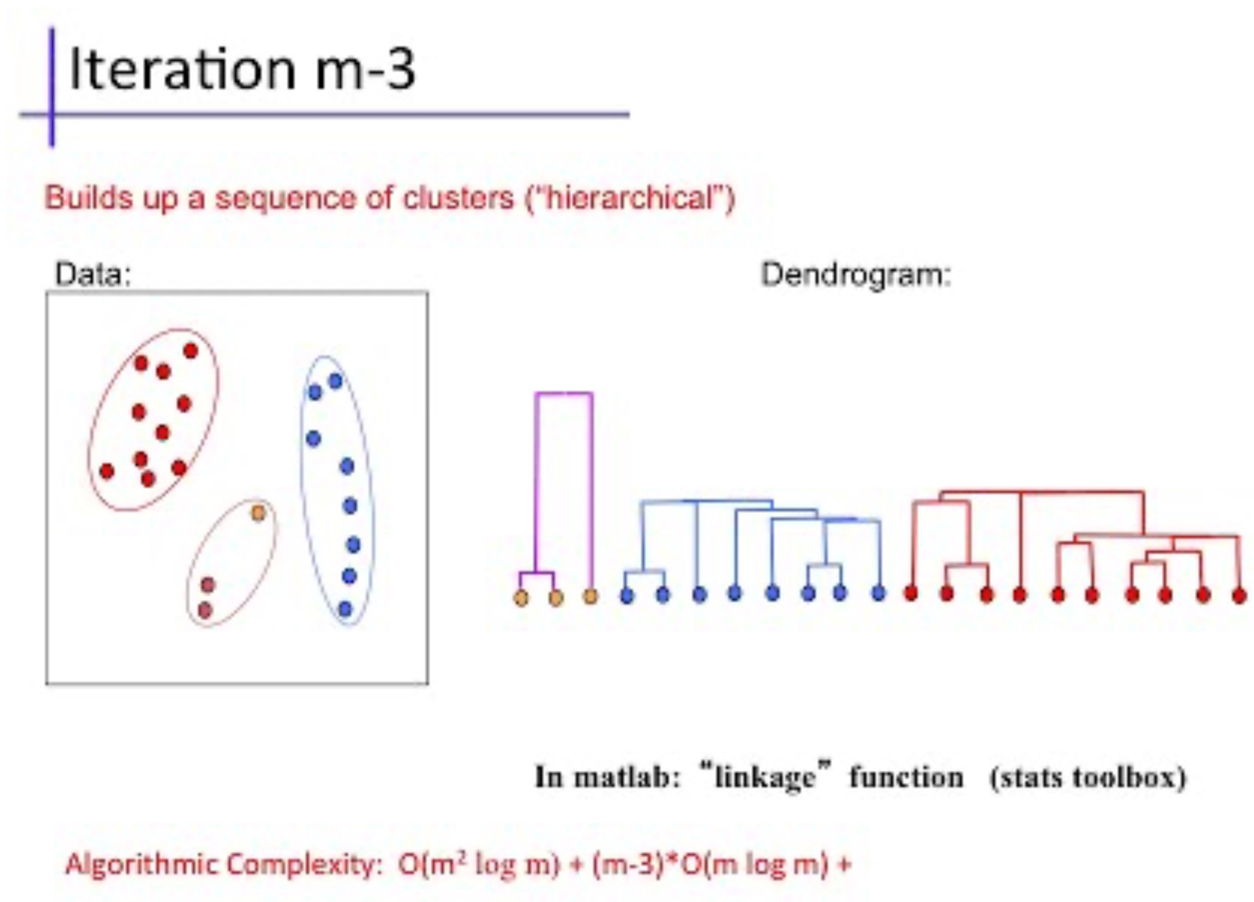
Projection of Documents in 3D Space



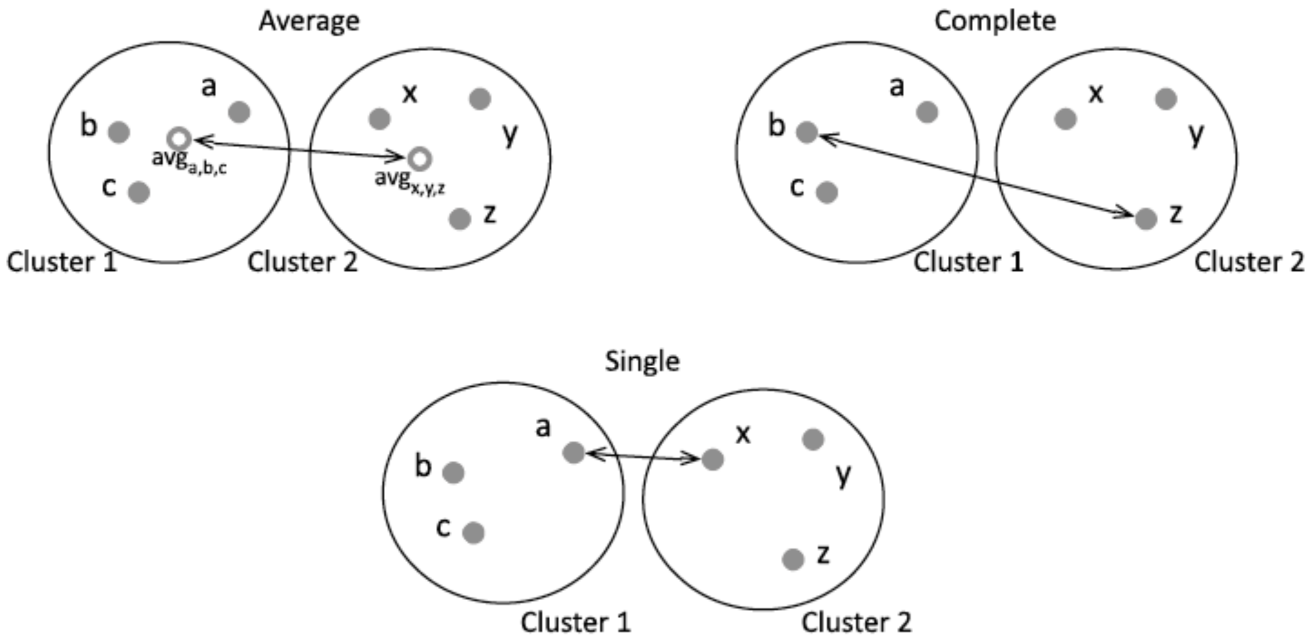
Agglomerative Hierarchical Clustering Algorithm



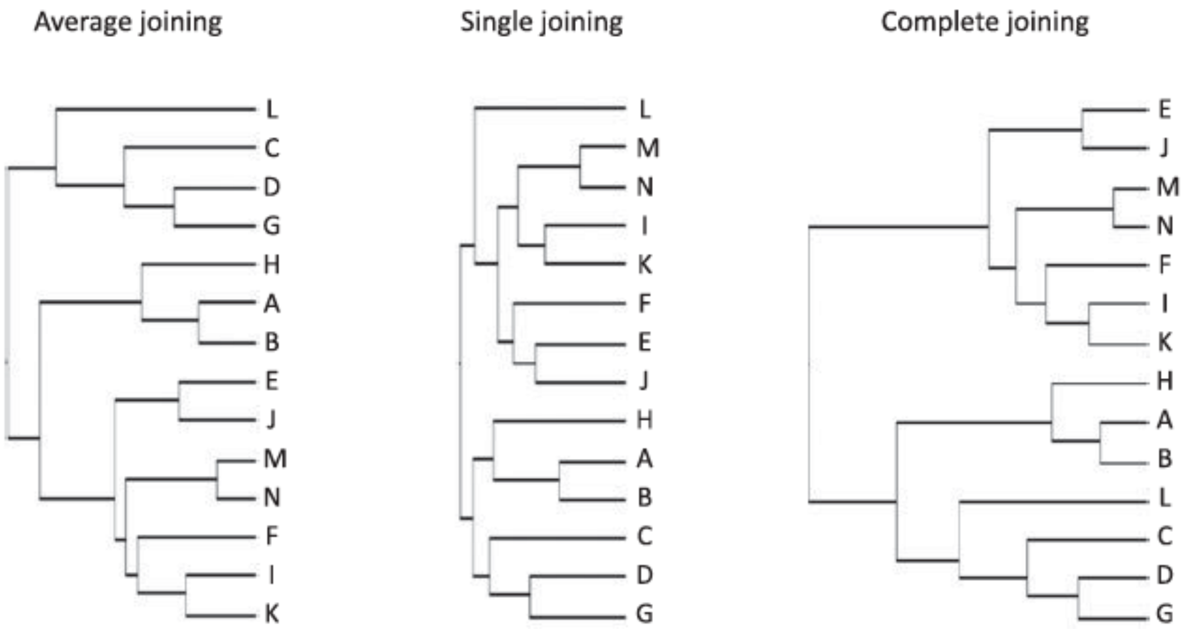
AHC Algorithm - Video



Linkage rules



AHC result


Average joining
L
C
D
G
H
A
B
E
J
M
N
F
I
K
Single joining
L
M
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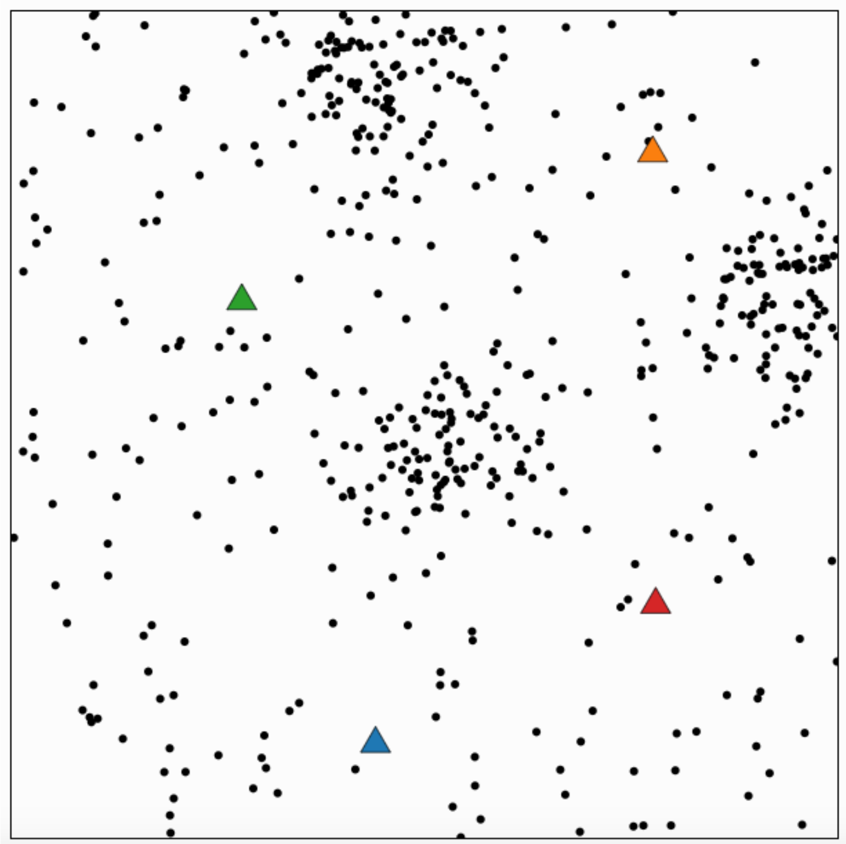
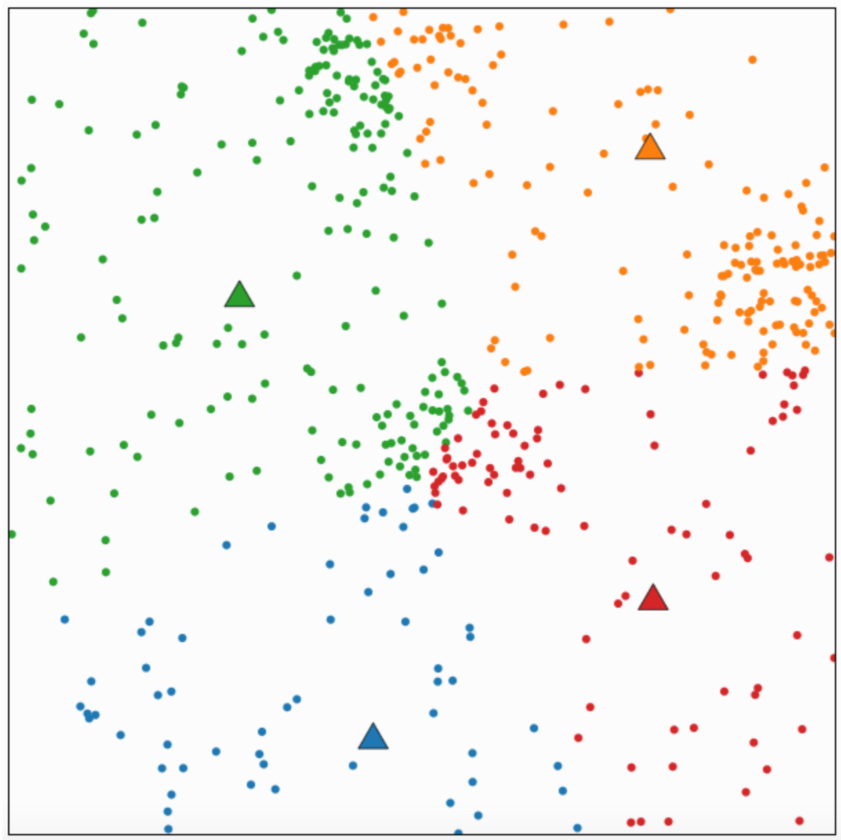
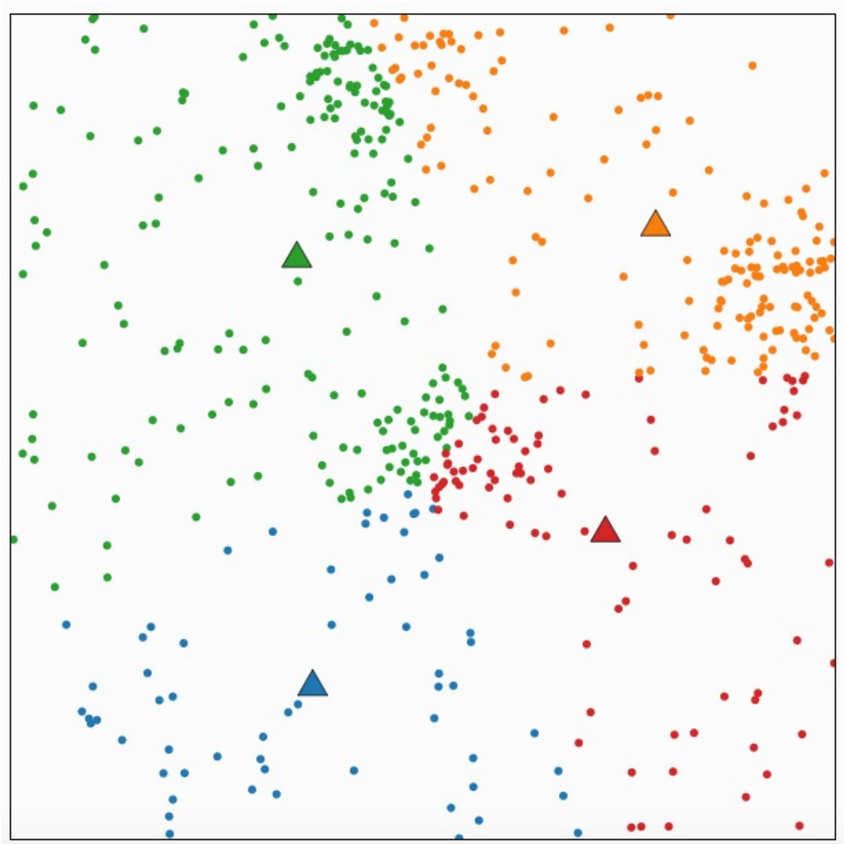
I
K
F
E
J
H
A
B
C
D
G
Complete joining
E
J
M
N
F
I
K
H
A
B
L
C
D
G

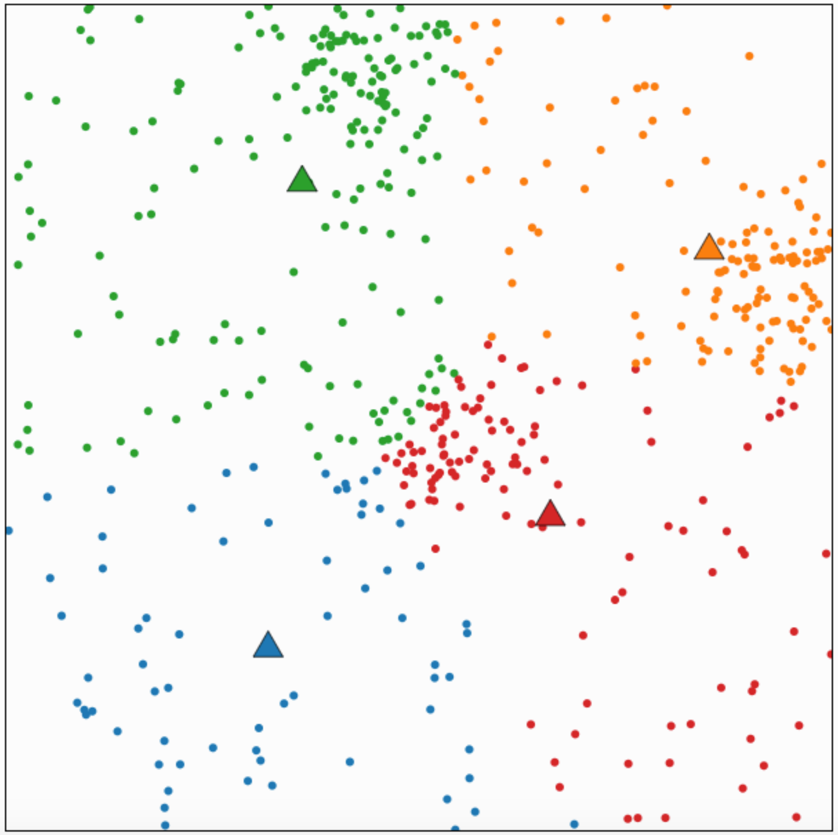
K-means clustering

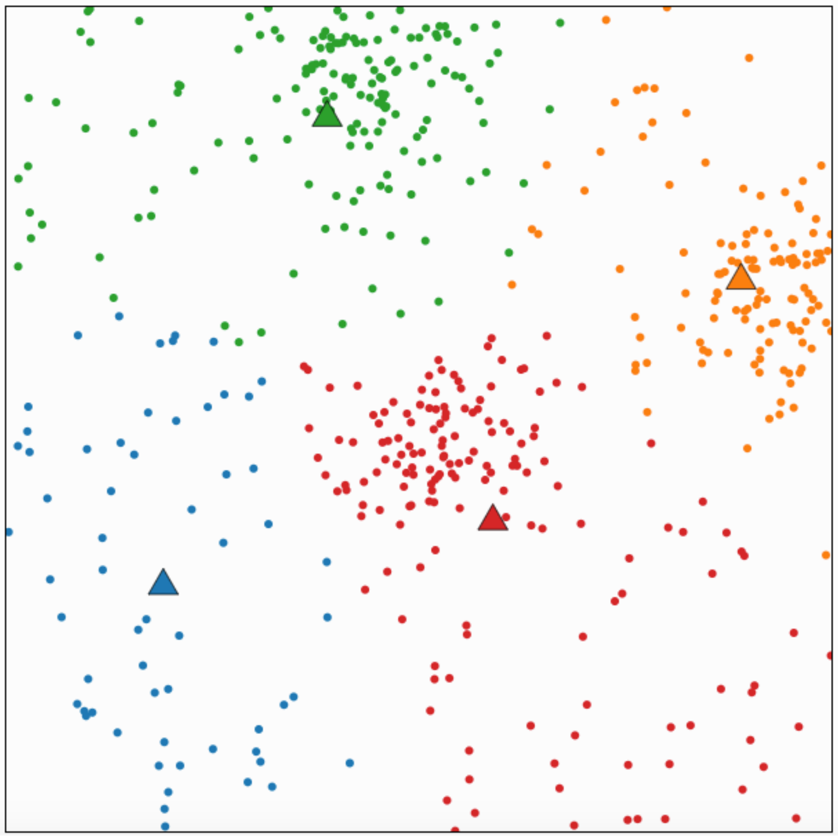
- A partitioning approach
  - Fix a number of clusters
  - Get “centroids” of each cluster
  - Assign things to closest centroid
  - Recalculate centroids
- Requires
  - A defined distance metric
  - A number of clusters
  - An initial guess as to cluster centroids
- Produces
  - Final estimate of cluster centroids
  - An assignment of each point to clusters



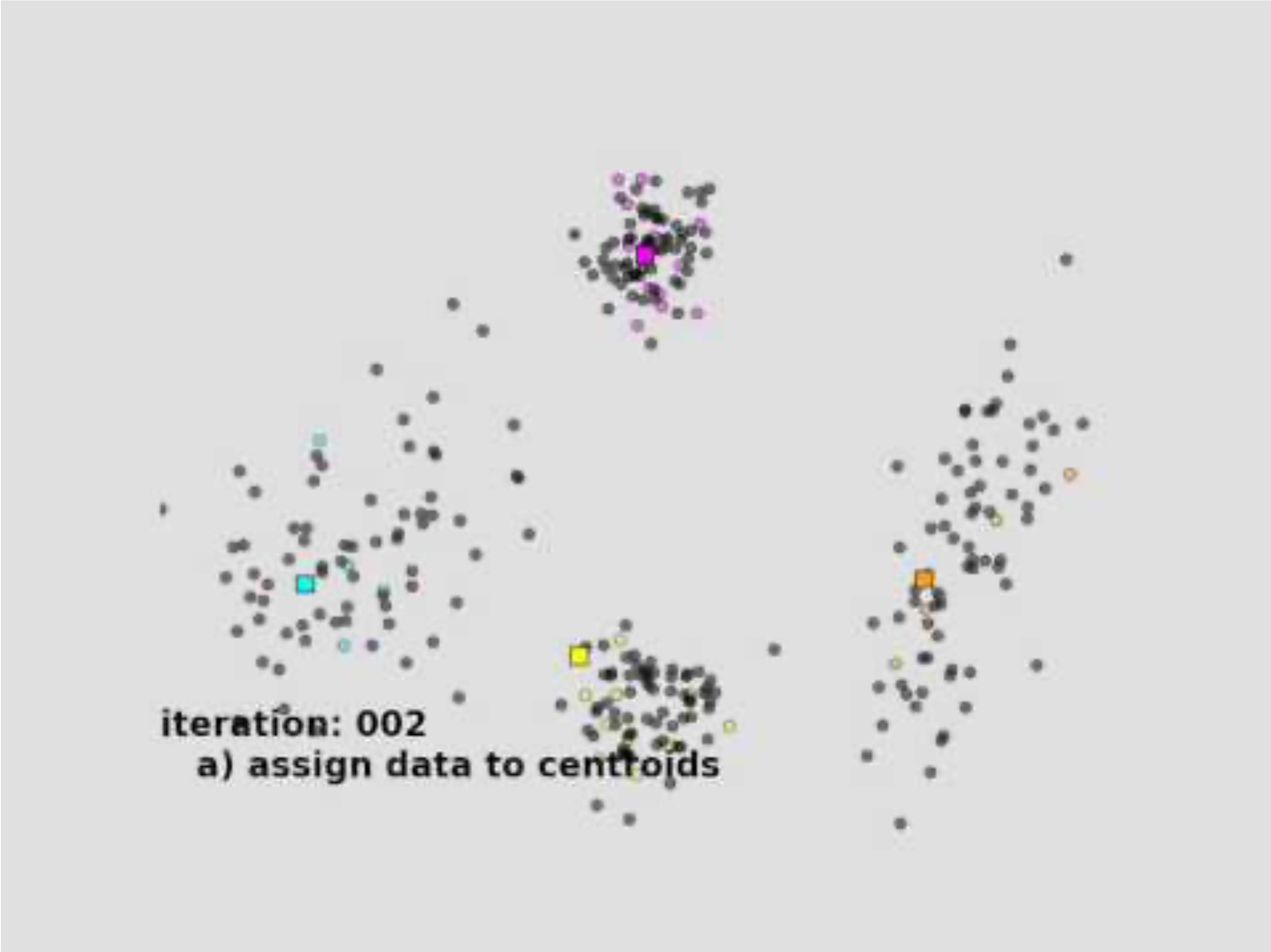








Kmeans Clustering Demo



Dimensionality reduction

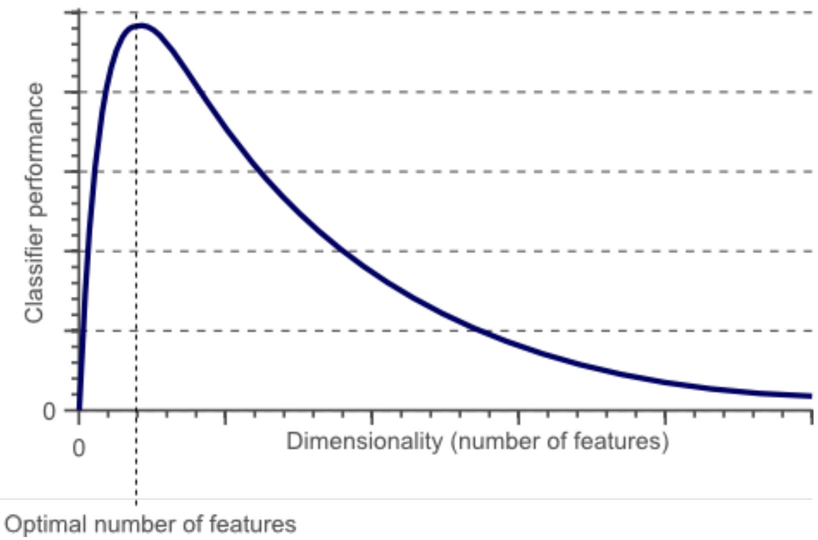
Principal Components Analysis and Feature Selection

Motivation

- Most machine learning and data mining techniques may not be effective for high-dimensional data
  - Curse of Dimensionality. Irrelevant and redundant features can “confuse” learners!
  - The intrinsic dimension may be small.

Curse of dimensionality

- The required number of samples (to achieve the same accuracy) grows exponentionally with the number of variables!
- In practice: number of training examples is fixed!
  - => the classifier's performance usually will degrade for a large number of features!

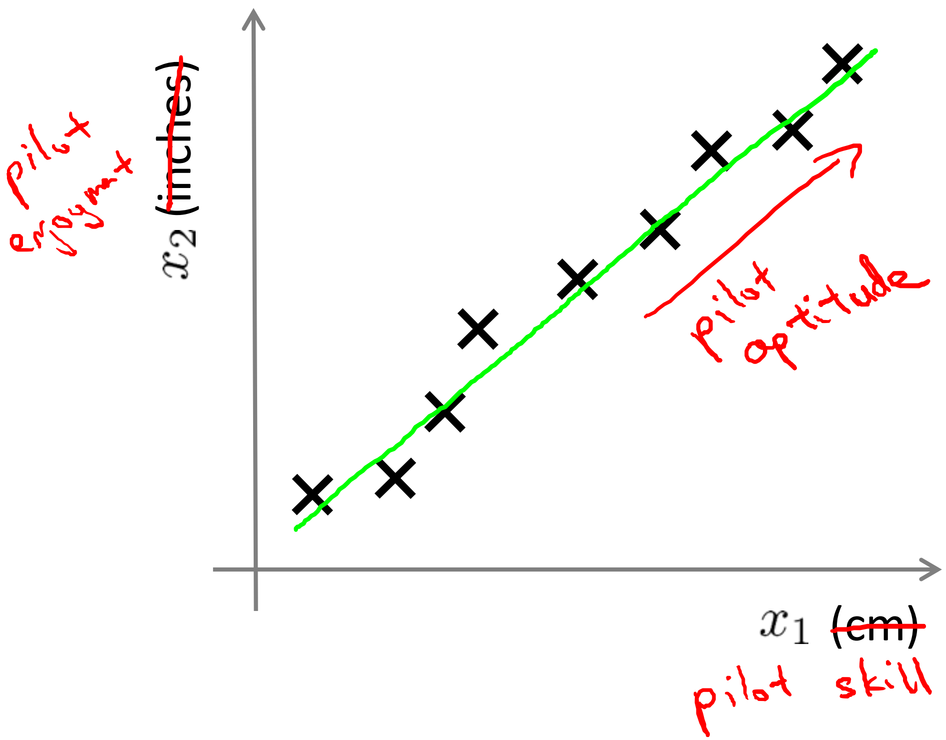


After a certain point, increasing the dimensionality of the problem by adding new features would actually degrade the performance of classifier.

Motivation

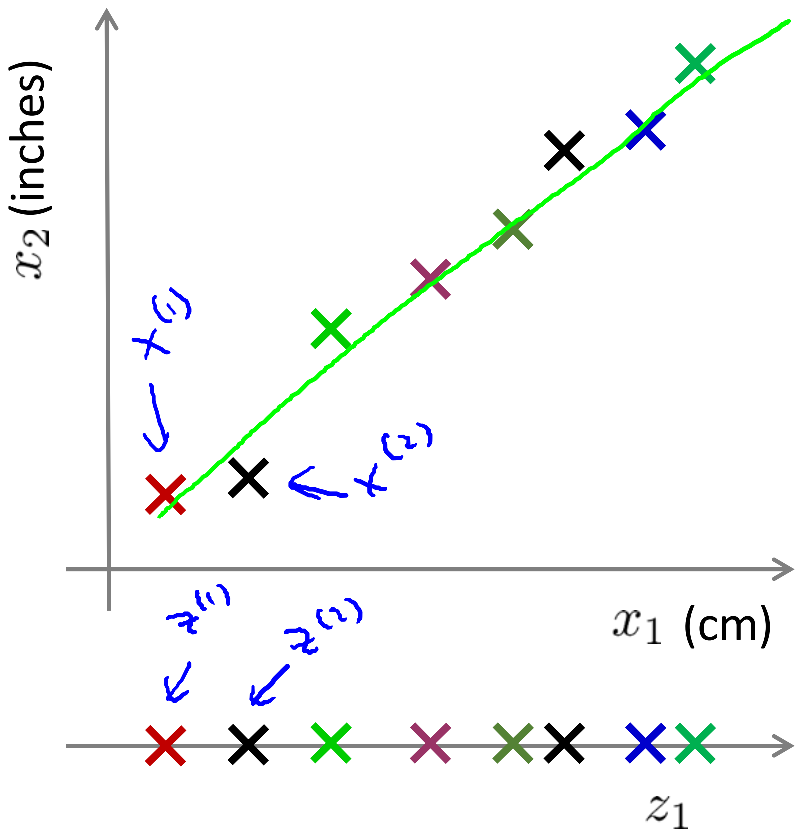
- Dimensionality reduction is an effective approach to downsizing data
  - Visualization: projection of high-dimensional data onto 2D or 3D.
  - Data compression: efficient storage and retrieval.
  - Noise removal: positive effect on query accuracy.

Data compression



Reduce data from 2D to 1D

Data compression (2)



Reduce data from 2D to 1D

$x^{(1)} \in \mathbb{R}^2 \rightarrow z^{(1)} \in \mathbb{R}$

$x^{(2)} \in \mathbb{R}^2 \rightarrow z^{(2)} \in \mathbb{R}$

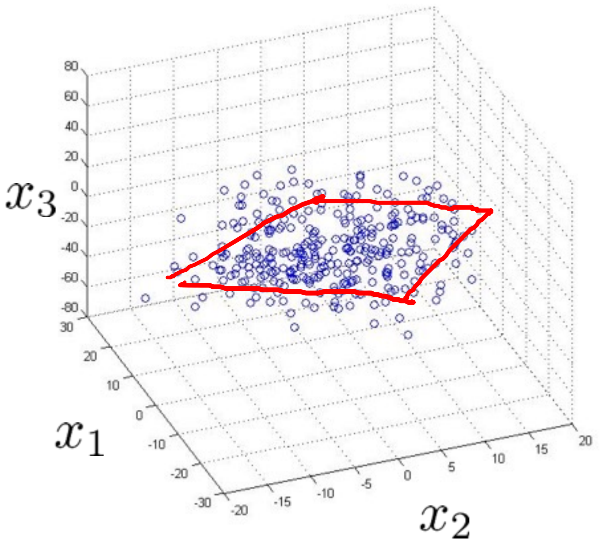
$\vdots$

$x^{(m)} \in \mathbb{R}^2 \rightarrow z^{(m)} \in \mathbb{R}$

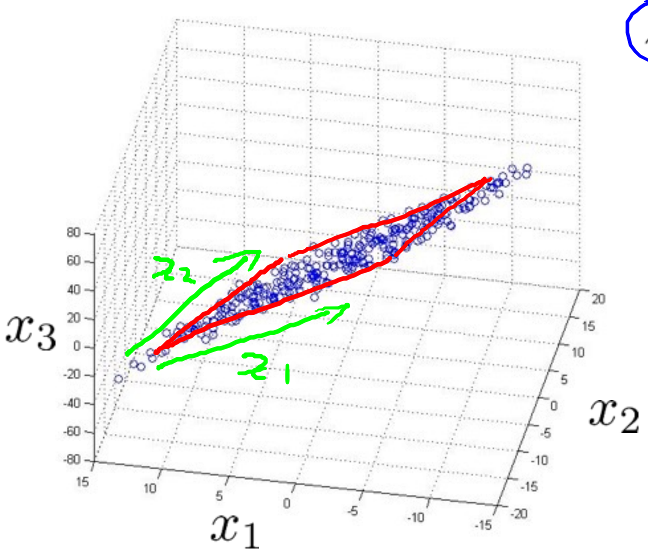
Data compression (2)

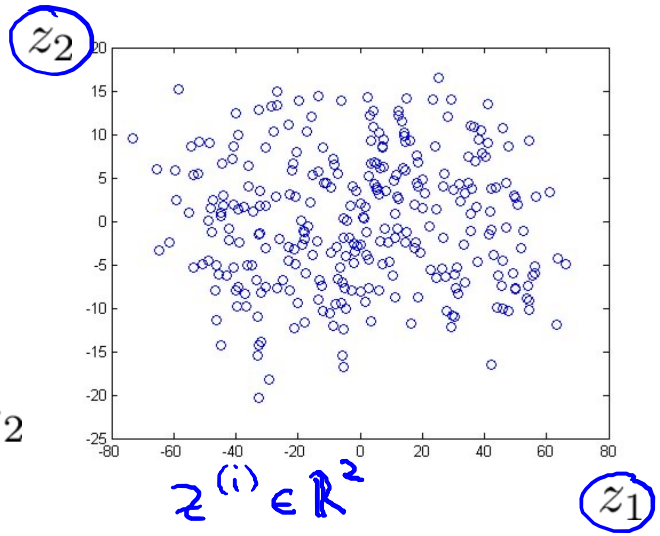
$1000D \rightarrow 100D$

Reduce data from 3D to 2D



$x^{(i)} \in \mathbb{R}^3$

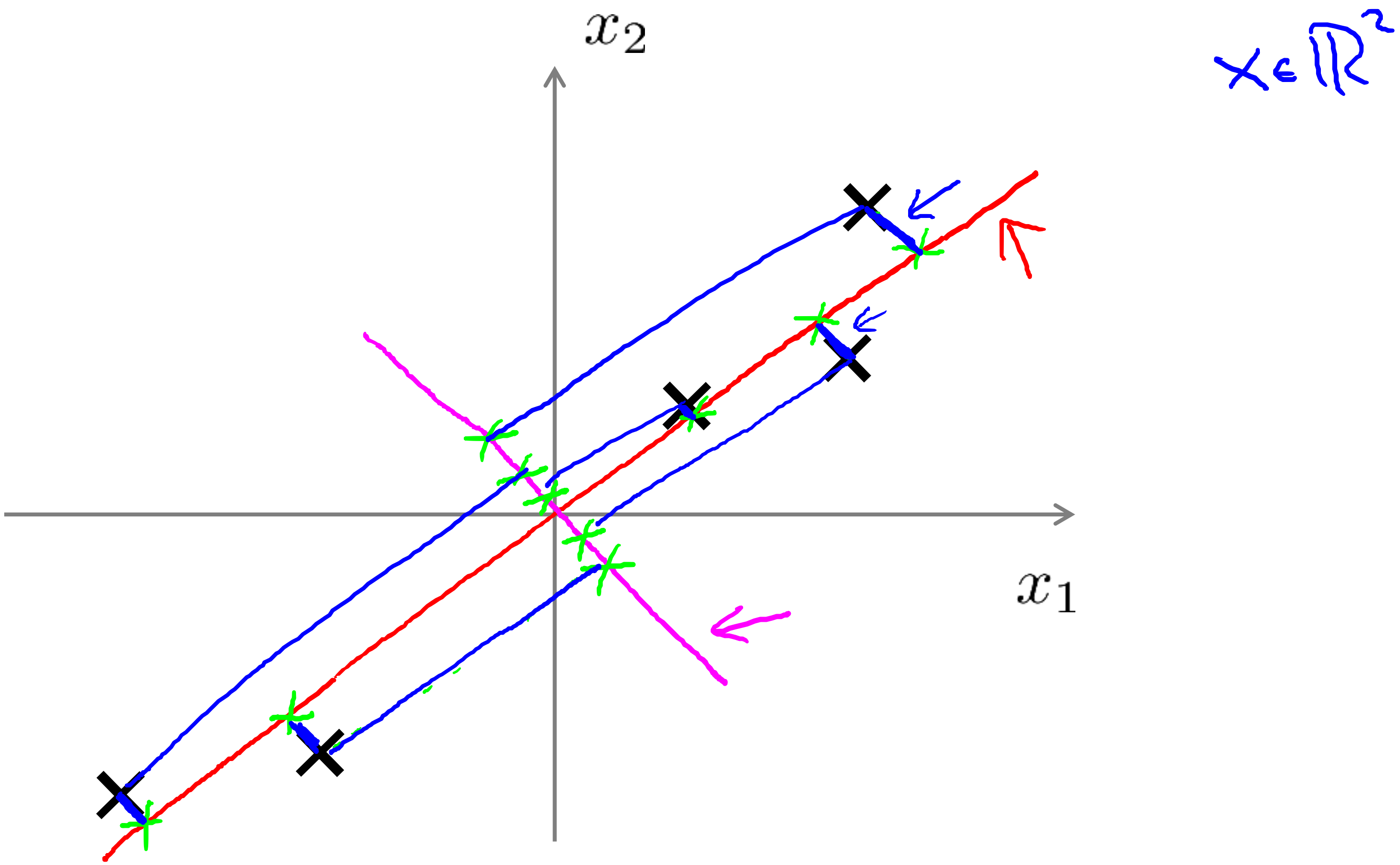




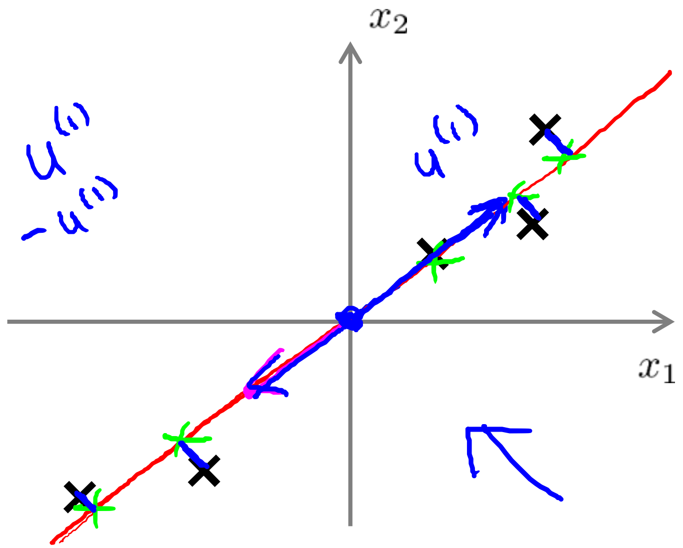
$z^{(i)} \in \mathbb{R}^2$

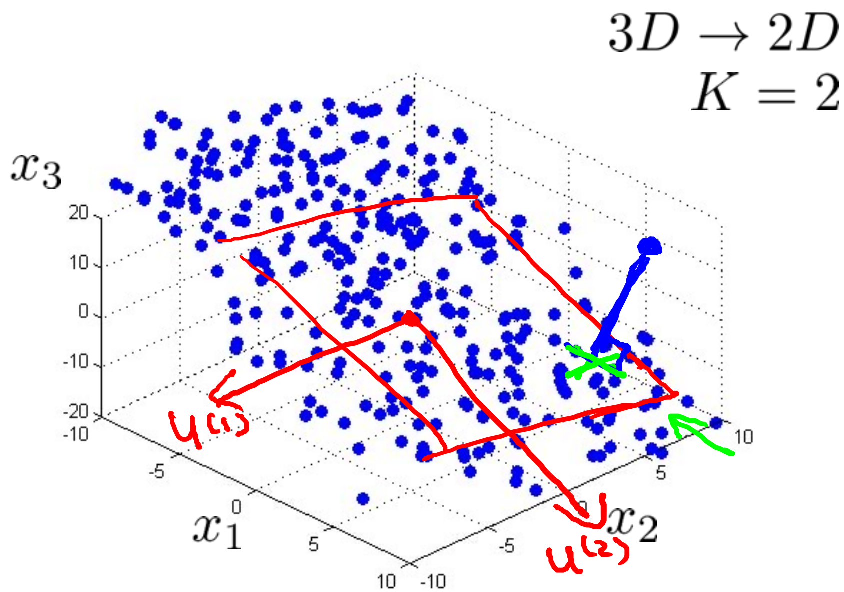
$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \qquad z^{(i)} = \begin{bmatrix} z_1^{(i)} \\ z_2^{(i)} \end{bmatrix}$

Principal Component Analysis (PCA) problem formulation



Principal Component Analysis (PCA) problem formulation





Reduce from 2-dimension to 1-dimension: Find a direction (a vector $u^{(1)} \in \mathbb{R}^n$) onto which to project the data so as to minimize the projection error.

Reduce from n-dimension to k-dimension: Find k vectors $u^{(1)}, u^{(2)}, \ldots, u^{(k)}$ onto which to project the data, so as to minimize the projection error.

Feature Selection

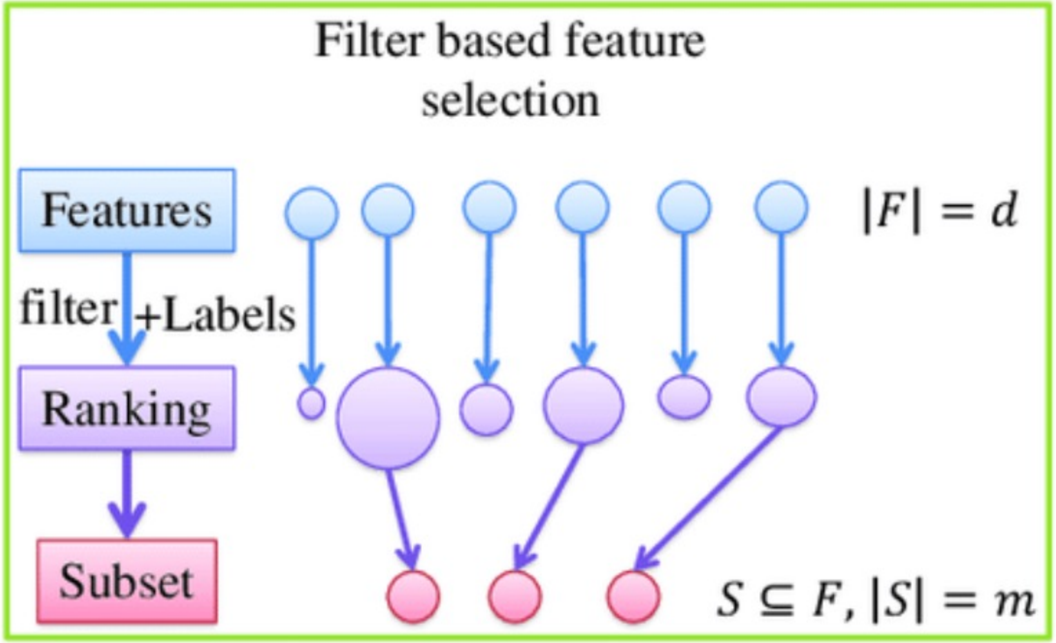
The role of feature selection in machine learning is,

- To reduce the dimensionality of feature space.
- To speed up a learning algorithm.
- To improve the predictive accuracy of a classification algorithm.
- To improve the comprehensibility of the learning results.

- 3 main approaches
  - Filter
  - Wrapper
  - Embedded

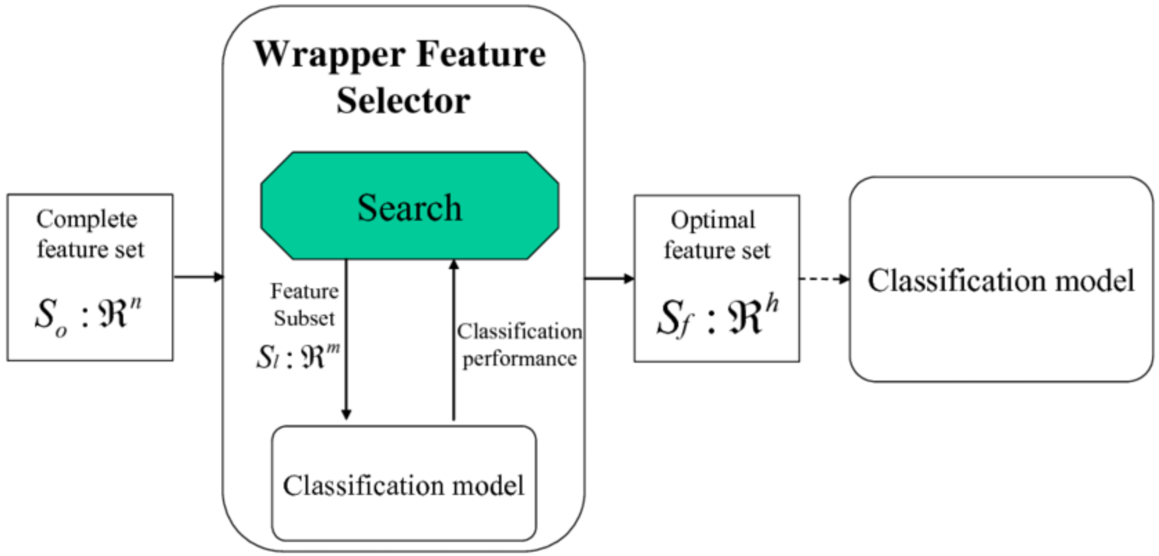
Filter-based Feature Selection

- Pre-processing step
- Prior to any classification models
- Ranking features based on their information score
    - Information Gain and its variants
    - Chi-square value
    - Fisher's score
    - Correlation Coefficient
    - and others.

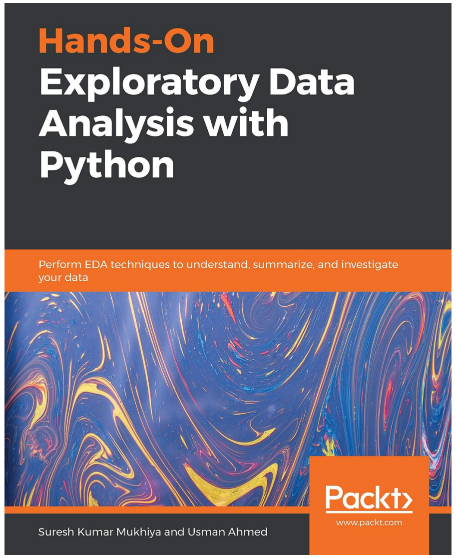


Wrapper-based Approach

- Select a sub-set of features based on the classification performance
- Rely on the target model
- … quite slow but sometimes better than Filter approaches



References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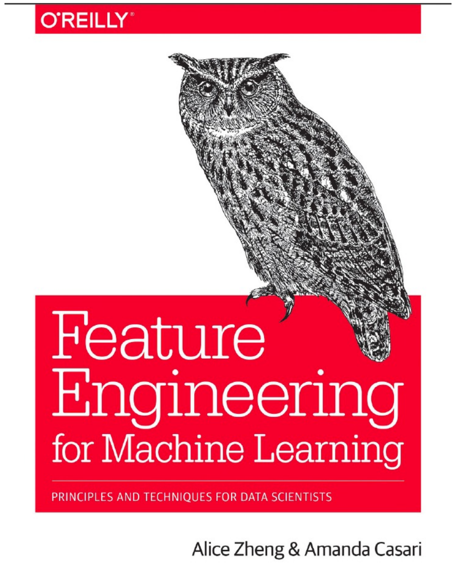
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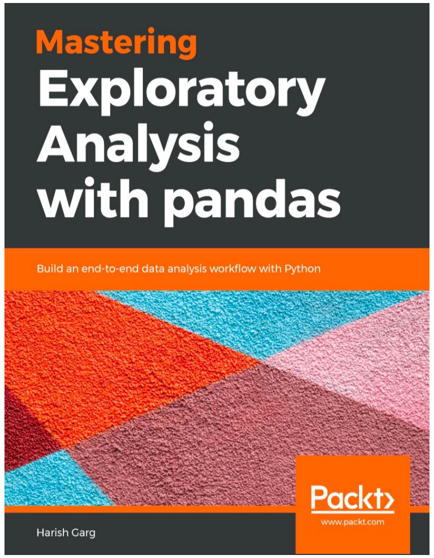
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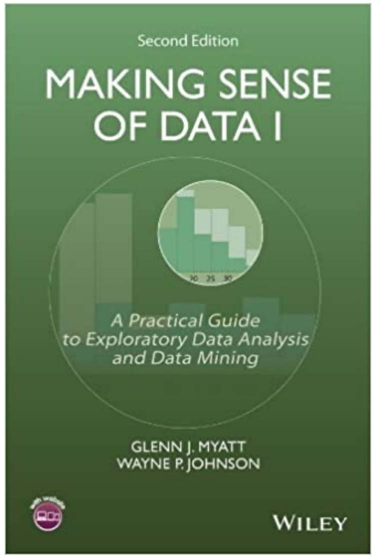
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WAYNE P. JOHNSON
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Thank you for your attention!!!

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Kmeans Clustering with Iris Dataset

Iris Dataset

In [2]:
```
iris = pd.read_csv("../input/iris-flower-dataset/IRIS.csv")
x = iris.iloc[:, [0, 1, 2, 3]].values
```

In [3]:
```
iris.info()
iris[0:10]
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

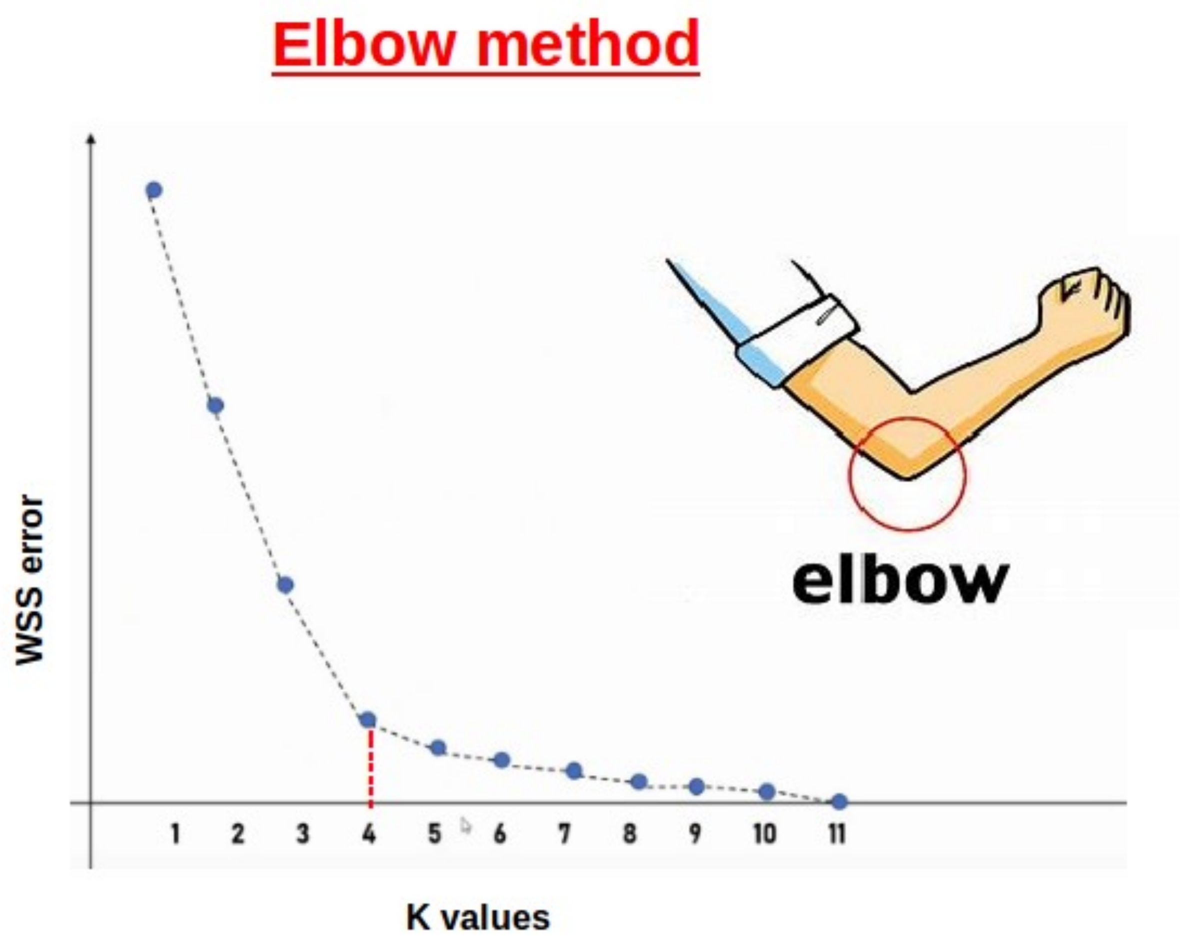

https://www.kaggle.com/datasets/arshid/iris-flower-dataset

How to apply K-Means Clustering?

- Choose the number of clusters k
- Select k random points from the data as centroids
- Assign all the points to the closest cluster centroid
- Recompute the centroids of newly formed clusters
- Repeat steps 3 and 4

How to choose k?

Elbow graph shows the within-cluster-sum-of-square (WCSS)



Elbow method for Iris data

In [10]:

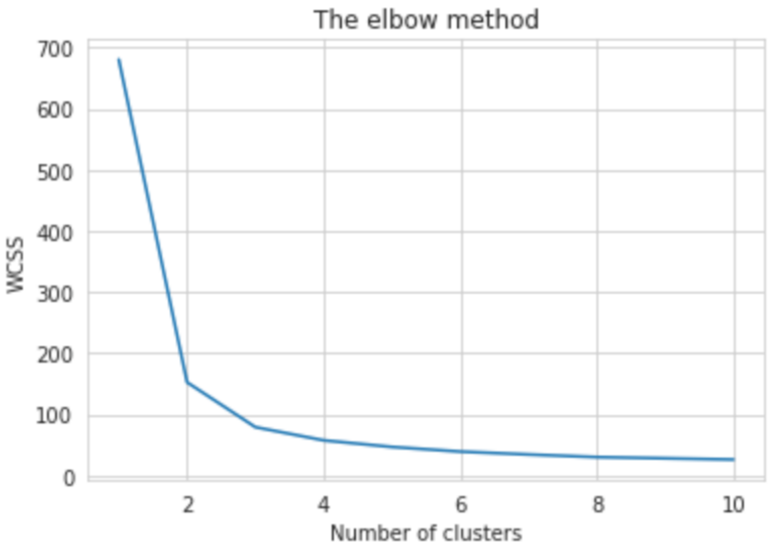
```
#Finding the optimum number of clusters for k-means classification
from sklearn.cluster import KMeans
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters = i, init = 'k-means++', max_iter = 300, n_init = 10, ran
dom_state = 0)
    kmeans.fit(x)
    wcss.append(kmeans.inertia_)
```

Elbow method for Iris data

In [11]:

```
plt.plot(range(1, 11), wcss)
plt.title('The elbow method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS') #within cluster sum of squares
plt.show()
```



K = 3

K-Means on Iris

In [12]:

```
kmeans = KMeans(n_clusters = 3, init = 'k-means++', max_iter = 300, n_init = 10, random_
state = 0)
y_kmeans = kmeans.fit_predict(x)
```

In [13]:

```
#Visualising the clusters
plt.scatter(x[y_kmeans == 0, 0], x[y_kmeans == 0, 1], s = 100, c = 'purple', label = 'Ir
is-setosa')
plt.scatter(x[y_kmeans == 1, 0], x[y_kmeans == 1, 1], s = 100, c = 'orange', label = 'Ir
is-versicolour')
plt.scatter(x[y_kmeans == 2, 0], x[y_kmeans == 2, 1], s = 100, c = 'green', label = 'Iri
s-virginica')

#Plotting the centroids of the clusters
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:,1], s = 100, c = 'r
ed', label = 'Centroids')

plt.legend()
```

K-Means on Iris